The **gap-closing estimand**

A causal approach
to study **interventions**
that **close disparities**
across social categories

Slides at
ianlundberg.org

Software at
ilundberg.github.io/gapclosing

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30 Jan 2025
Chinese University of Hong Kong

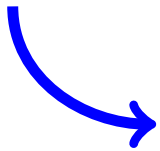
Paper in *Sociological Methods and Research*. Replication code on Dataverse. R package `gapclosing` on CRAN. Research reported in this publication was supported by The Eunice Kennedy Shriver National Institute of Child Health & Human Development of the National Institutes of Health under Award Number P2CHD047879 and by the National Science Foundation under Award Number 2104607.

$$
\begin{aligned}
Y = \beta_0 &+ \beta_1(\text{Income}) \\
&+ \beta_2(\text{Race} = \text{Hispanic}) \\
&+ \beta_3(\text{Race} = \text{Non-Hispanic Black}) \\
&+ \beta_4(\text{Race} = \text{Other Non-Hispanic}) \\
&+ \beta_5(\text{College Degree}) \\
&+ \beta_6(\text{Two-parent household}) \\
&+ \epsilon
\end{aligned}
$$

What is your estimand?

Lundberg, Johnson, Stewart
What is Your Estimand?
American Sociological Review 2021

What is your estimand?

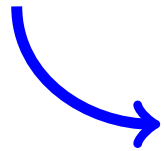


The **purpose** of the statistical analysis

stated **outside** of the model

Lundberg, Johnson, Stewart
What is Your Estimand?
American Sociological Review 2021

What is your estimand?



The **purpose** of the statistical analysis

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Y_i

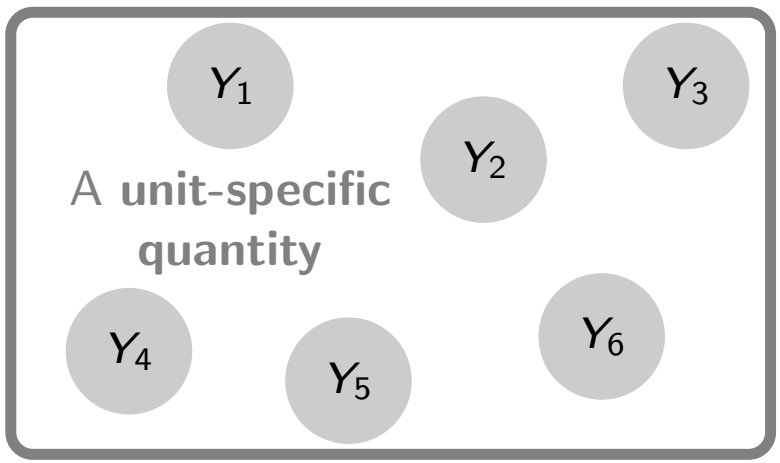
A **unit-specific quantity**

Lundberg, Johnson, Stewart
What is Your Estimand?
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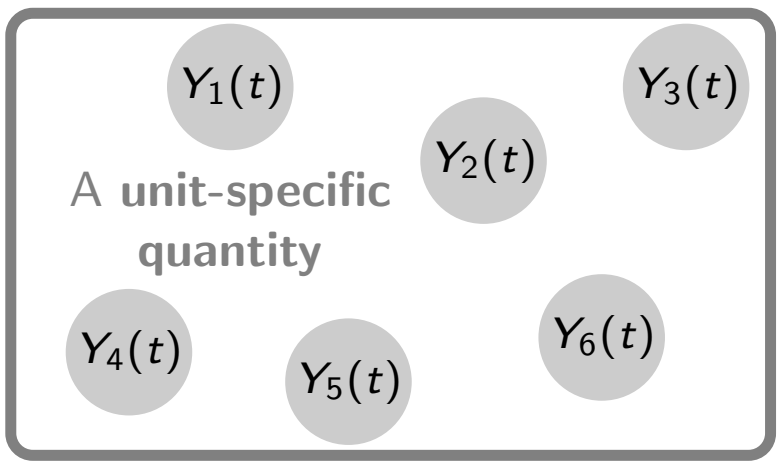
Aggregated over a **target population**

Lundberg, Johnson, Stewart
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The **purpose** of the statistical analysis

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Aggregated over a **target population**

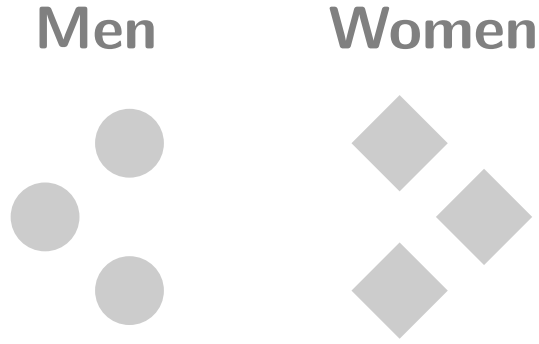
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The Gap-Closing Estimand:
A Causal Approach to Study Interventions That Close Disparities Across Social Categories

Ian Lundberg (UCLA)

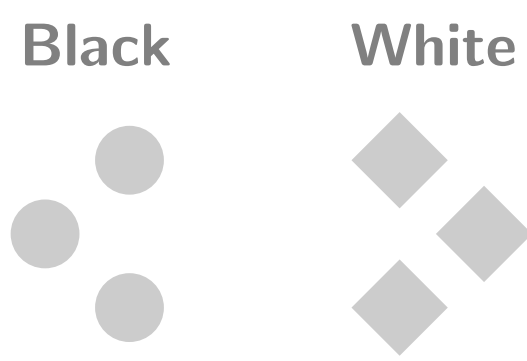
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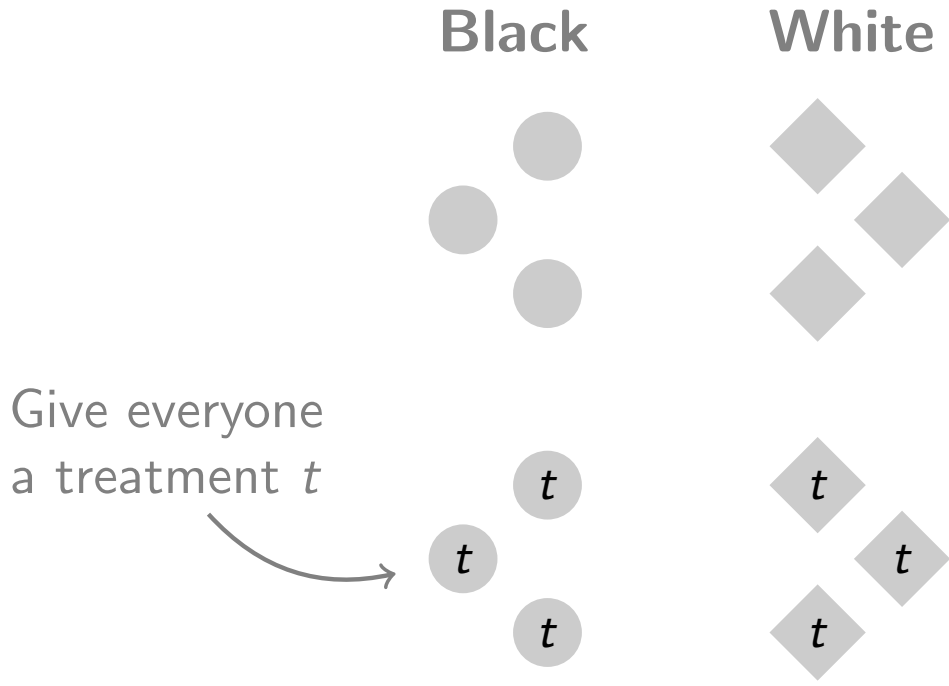
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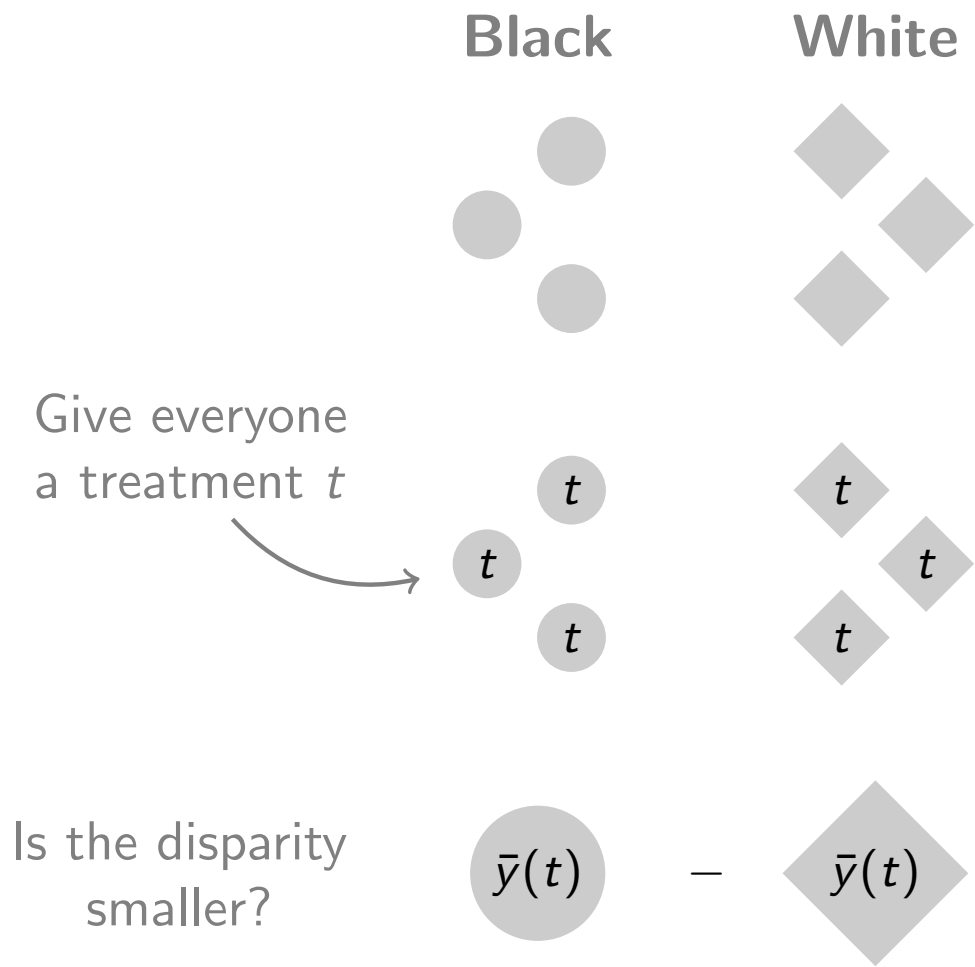
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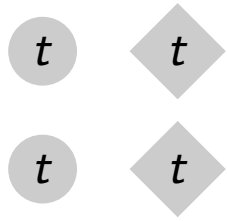
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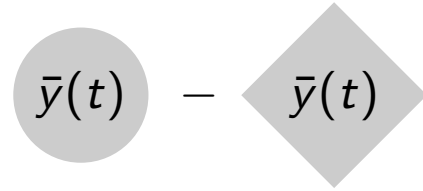
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Categories

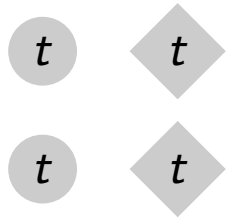


Treatment

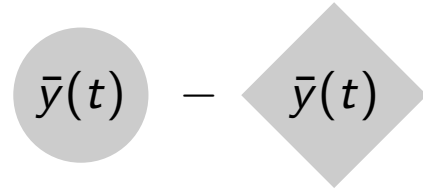


Counterfactual Disparity

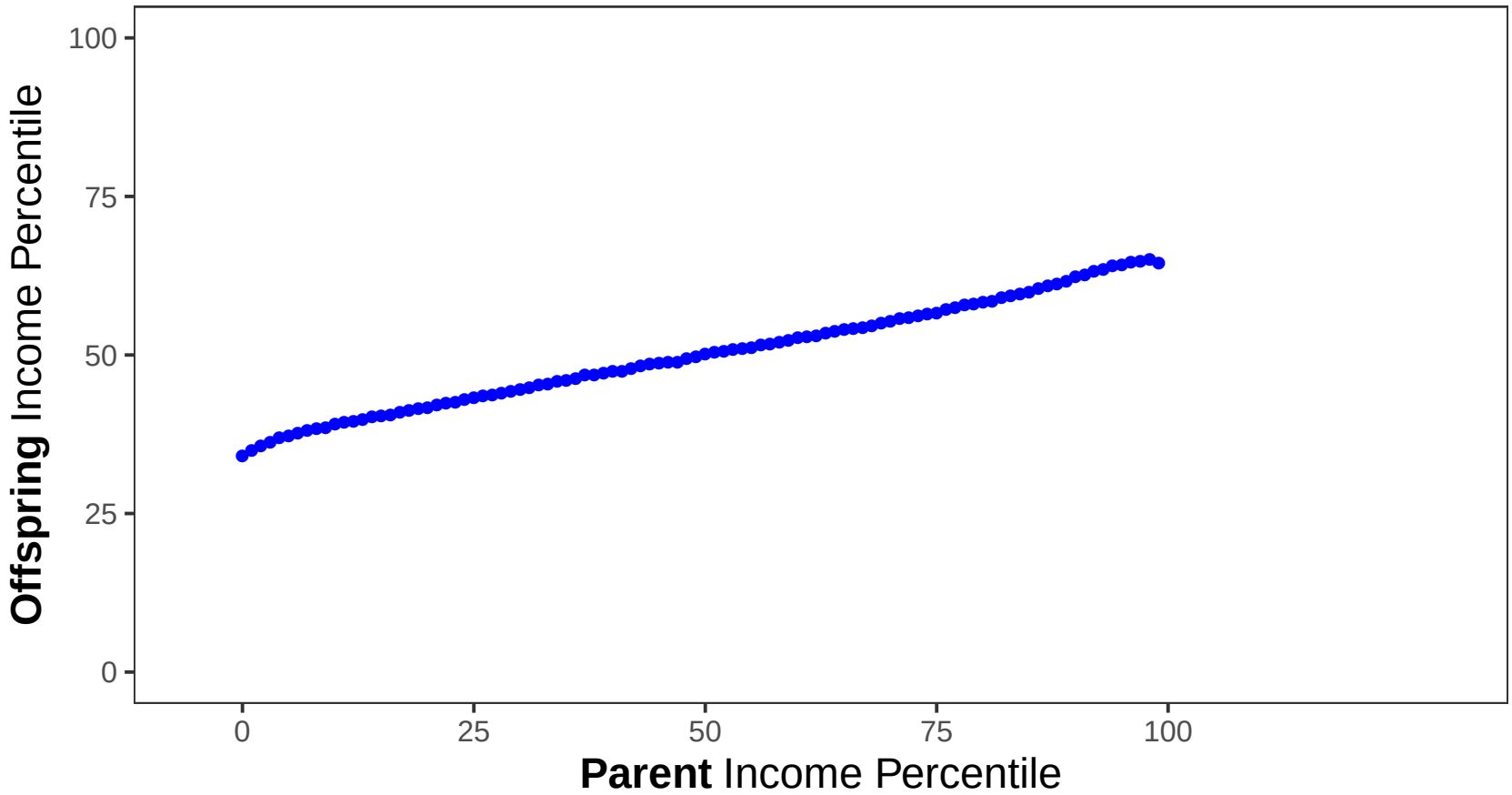
Categories



Treatment

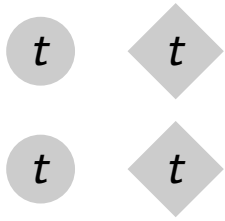


Counterfactual Disparity

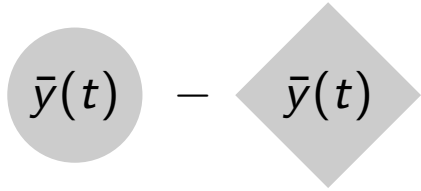


Chetty et al. 2017

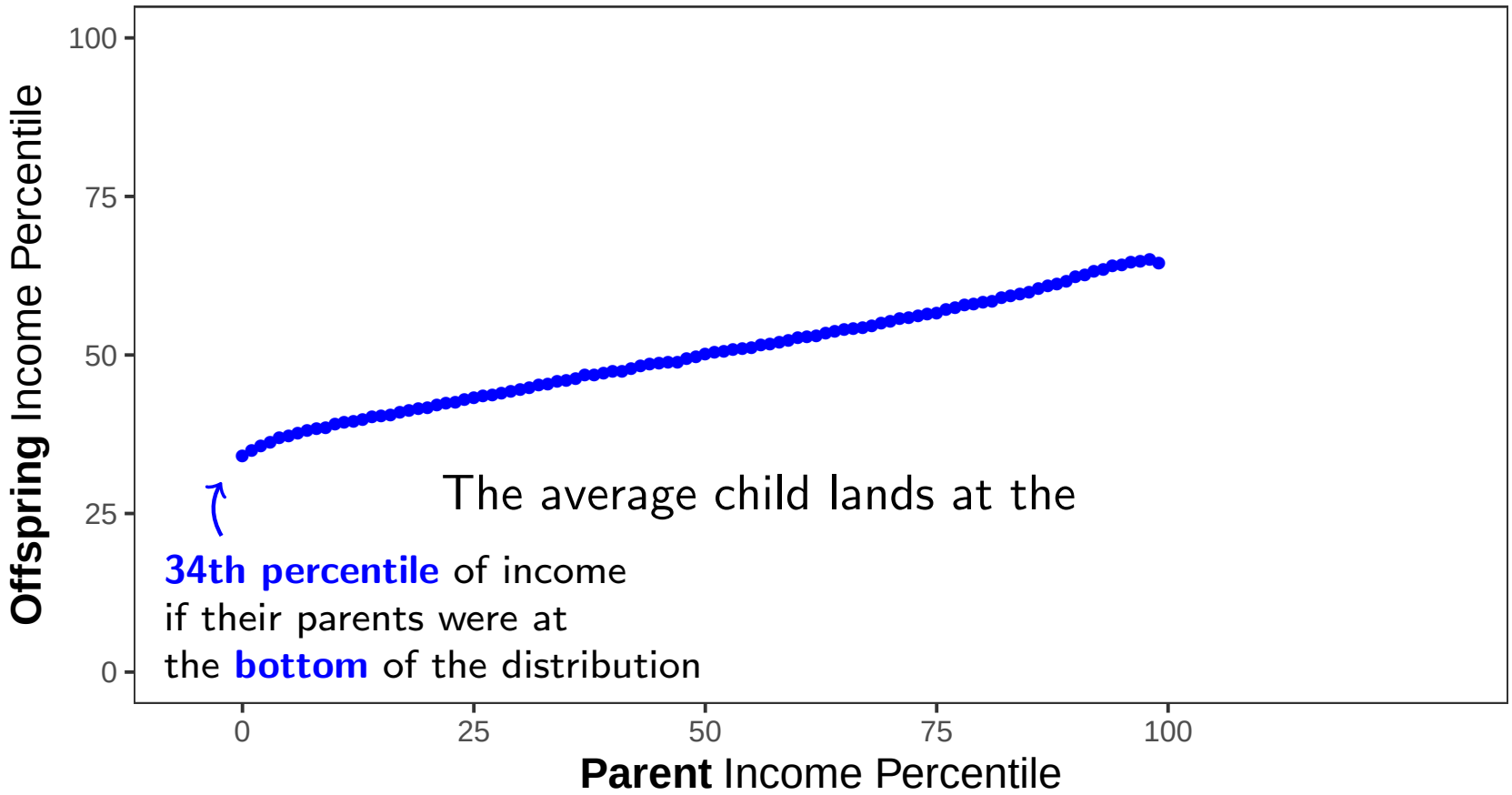
Categories



Treatment

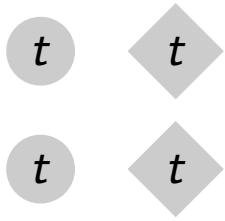


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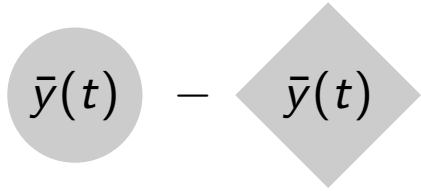


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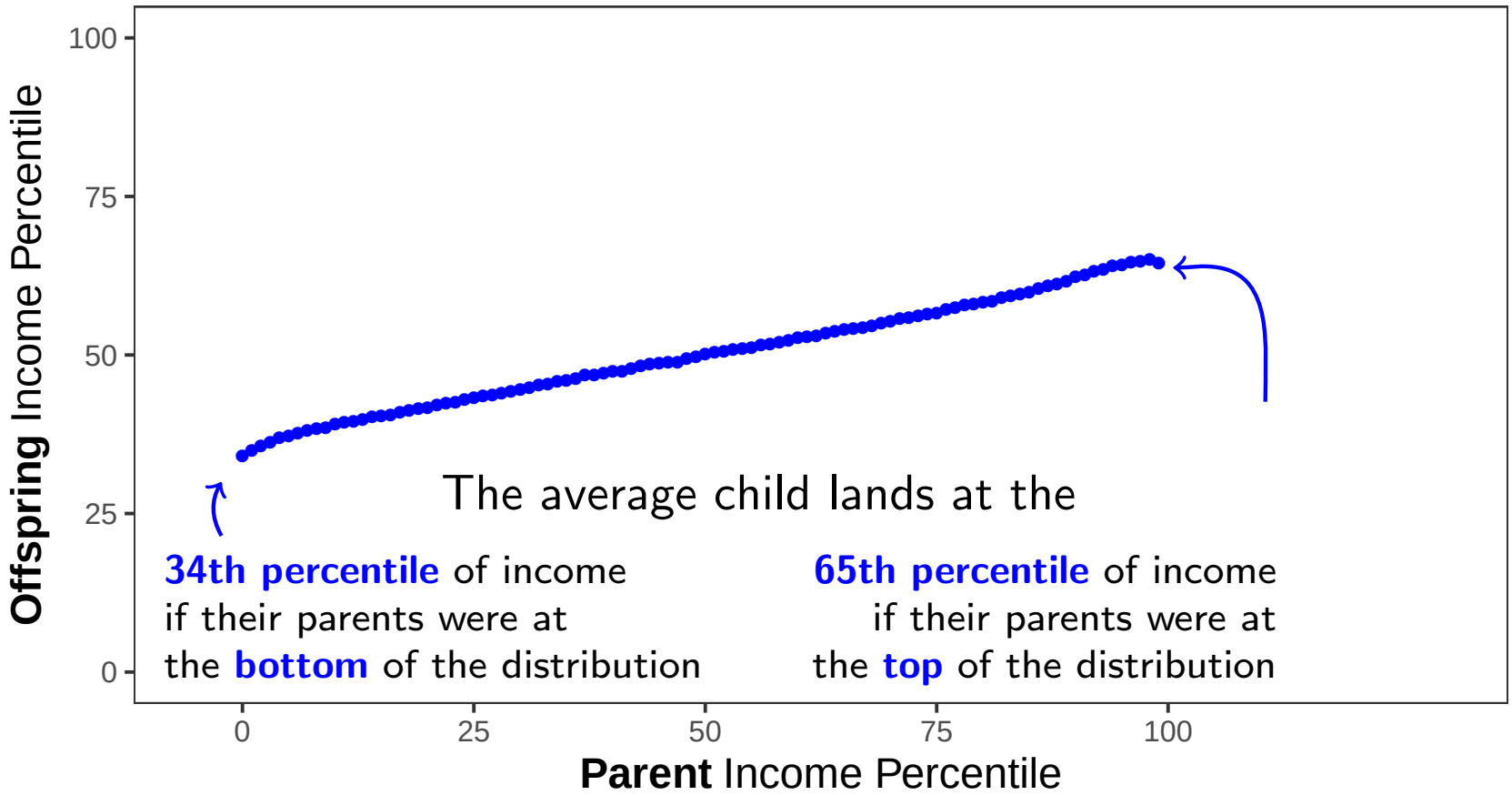
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Treatment

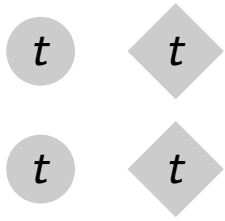


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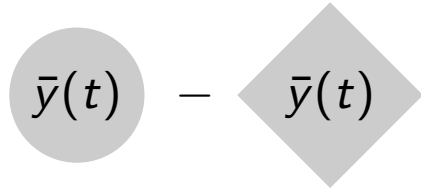


Chetty et al. 2017

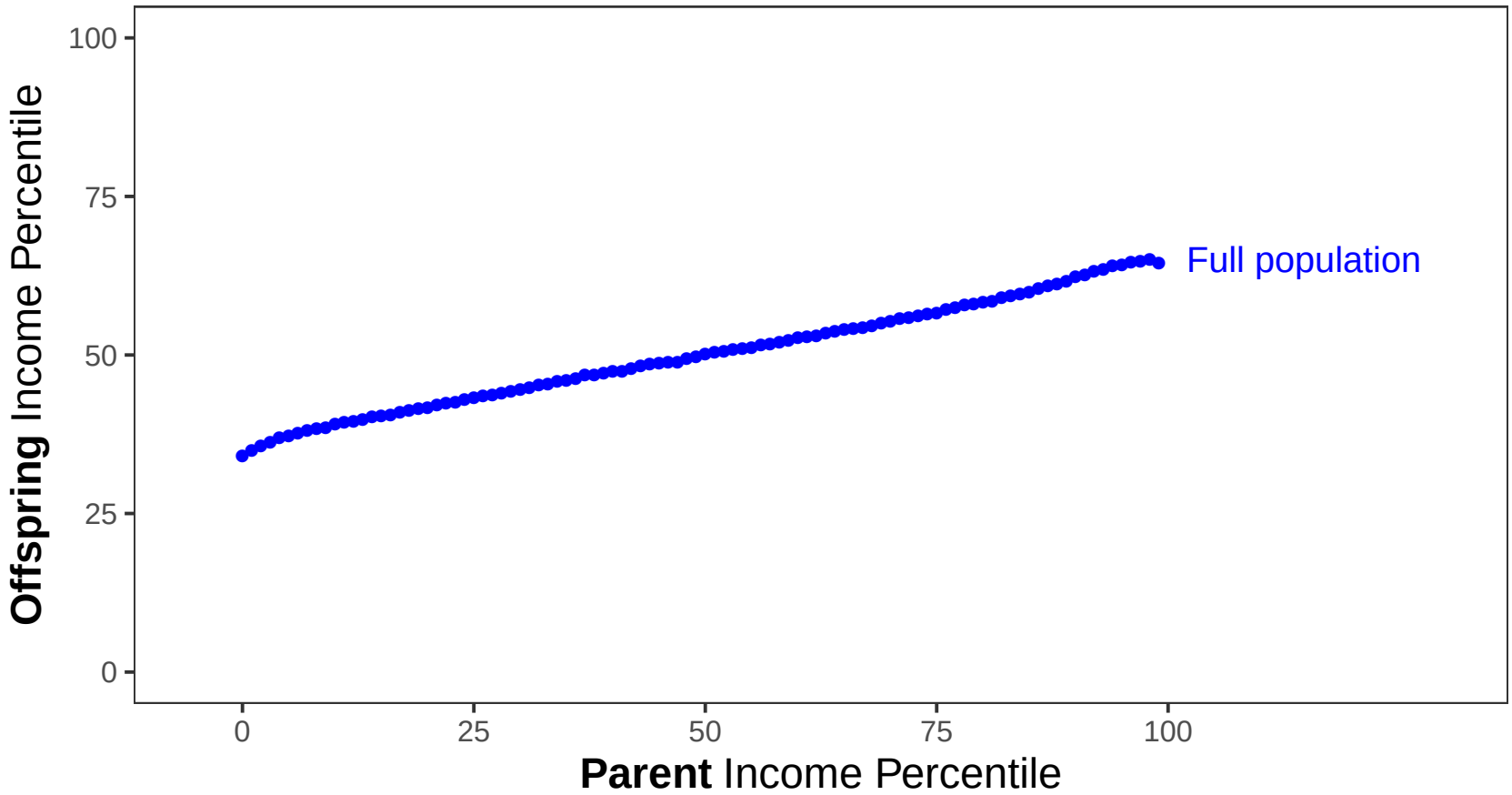
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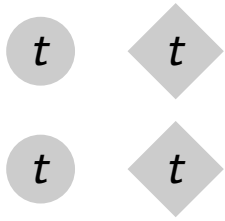


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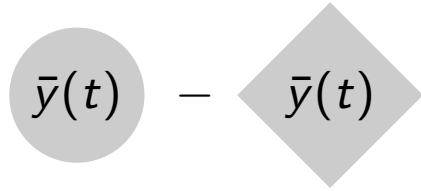


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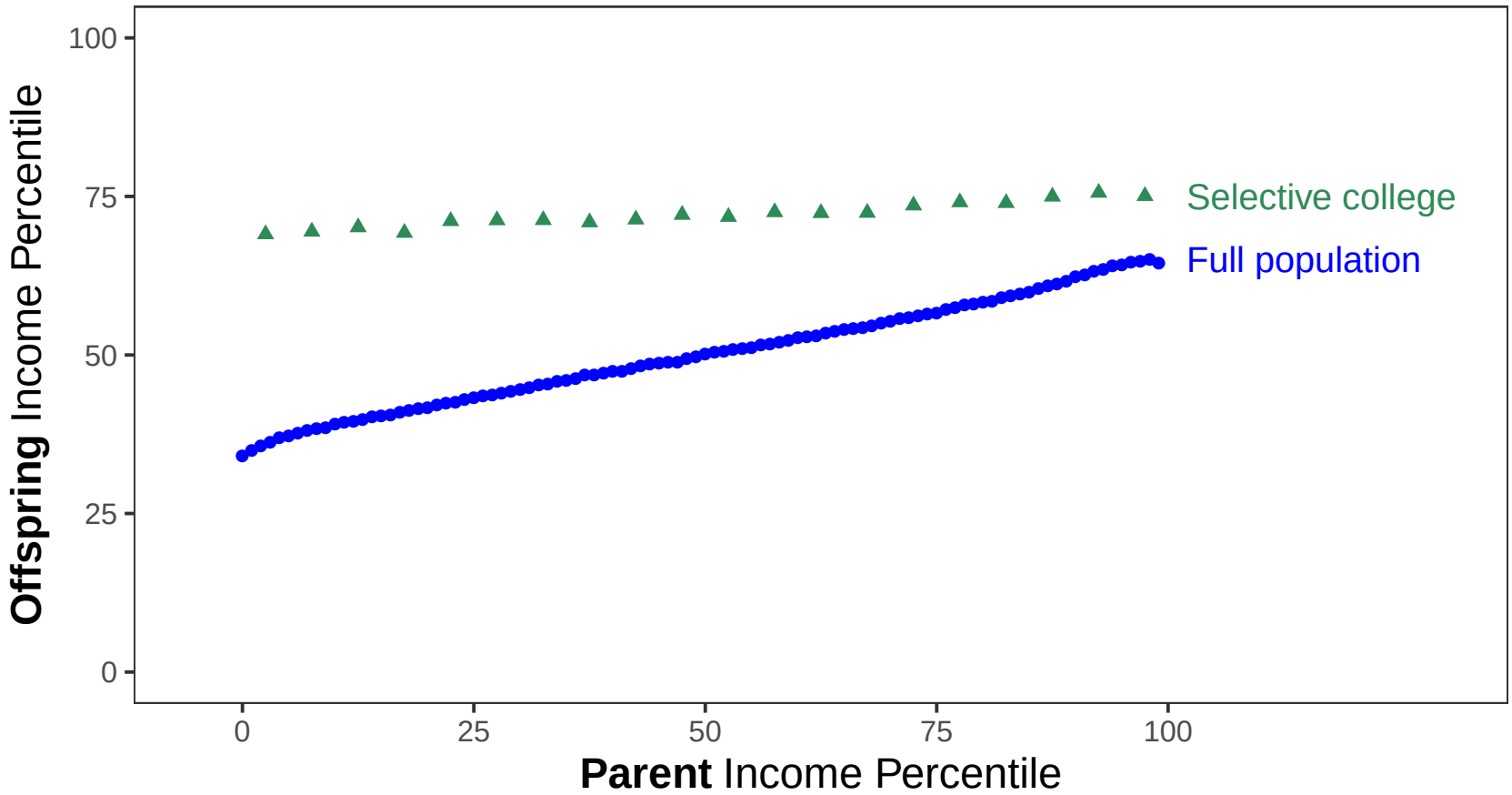
Categories



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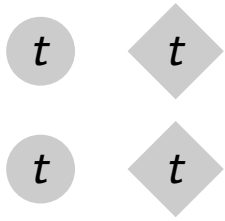


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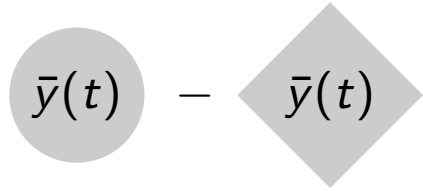


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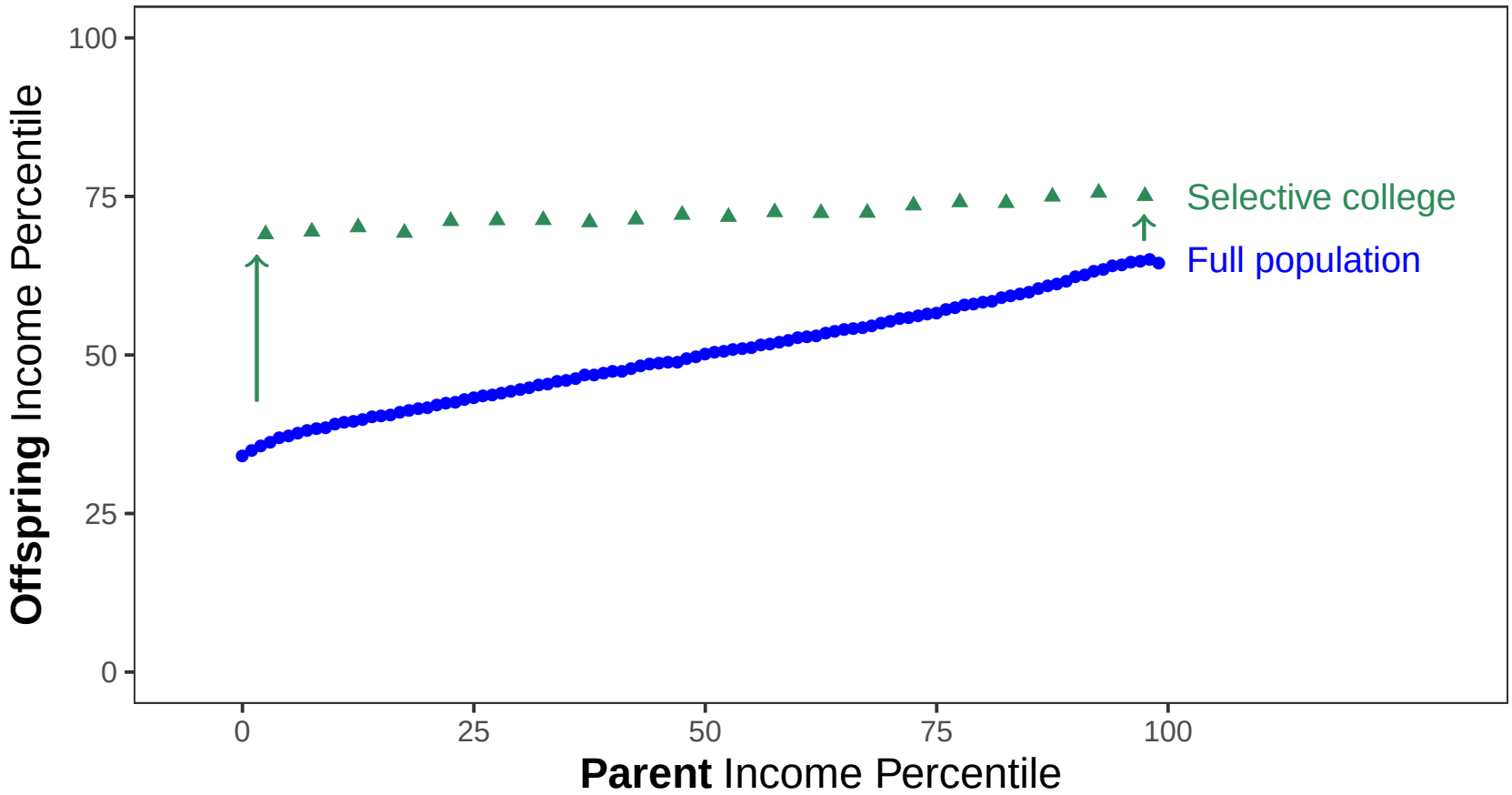
Categories



Treatment

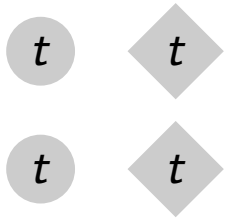


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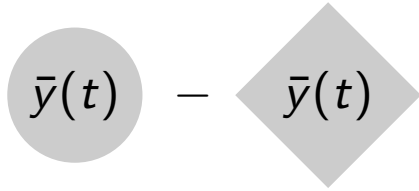


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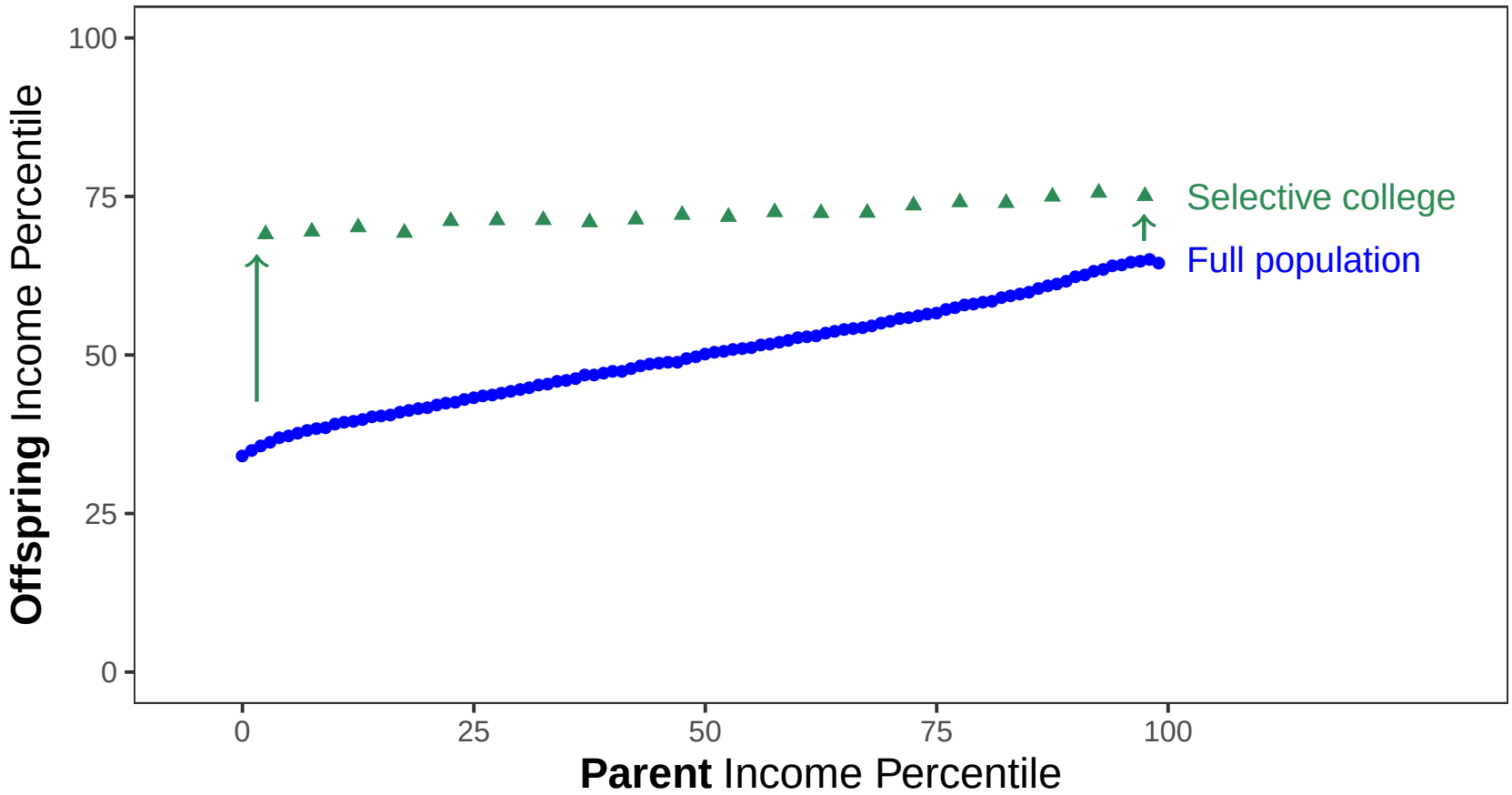
Categories
Parent Income



Treatment
Selective College



Counterfactual Disparity
Offspring Income



Chetty et al. 2017

The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention?
- Identify the estimand
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- Produce software
- Impact: Others using these ideas

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  - what is the treatment?
  - what is the scope of the intervention? (contribution)
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  - by weighting
  - by doubly robust estimation (contribution)
- Produce software (contribution)
- Impact: Others using these ideas

Past work: Causal decomposition analysis

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Unadjusted

$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted

$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

Past work: Causal decomposition analysis

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Effect of race

Past work: Causal decomposition analysis

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× Effect of race

Vanderweele & Robinson 2014

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Choice of intervention targets — Jackson & Vanderweele 2018

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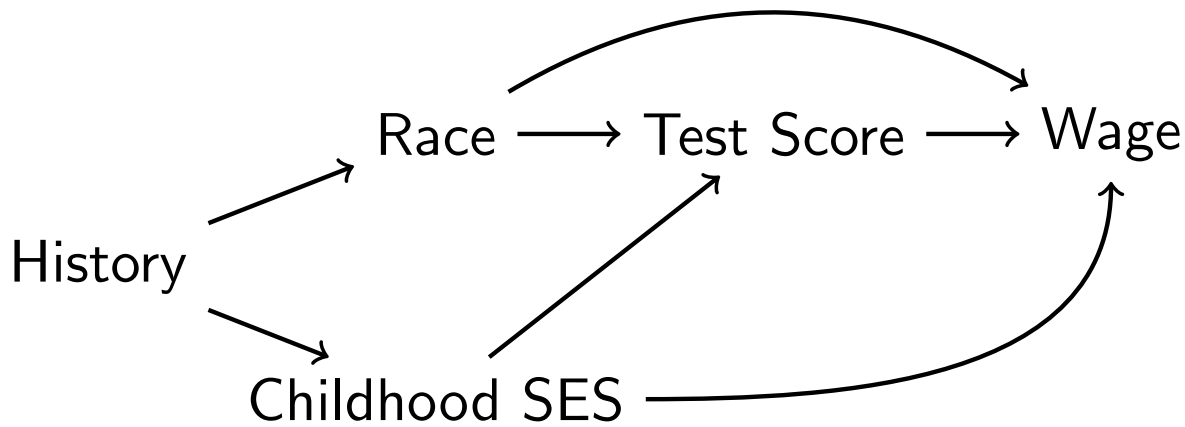
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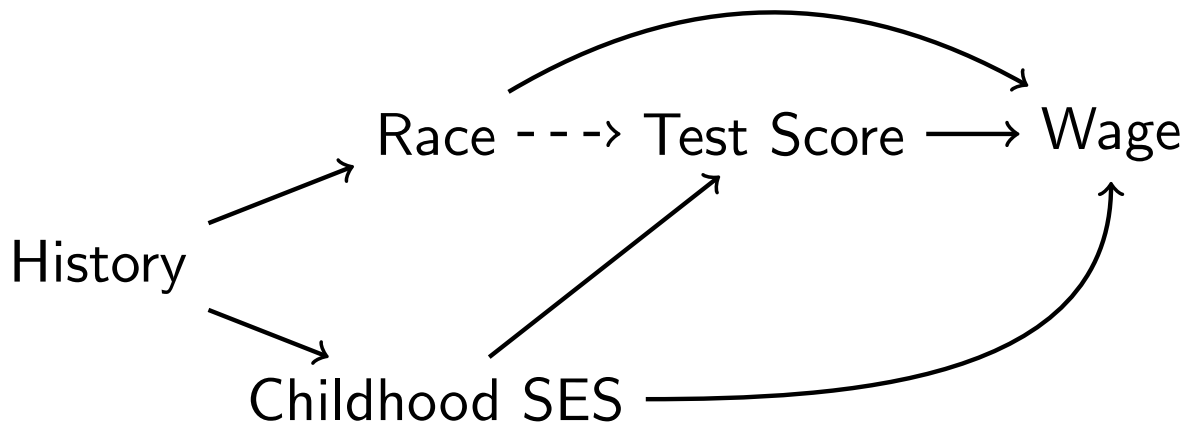
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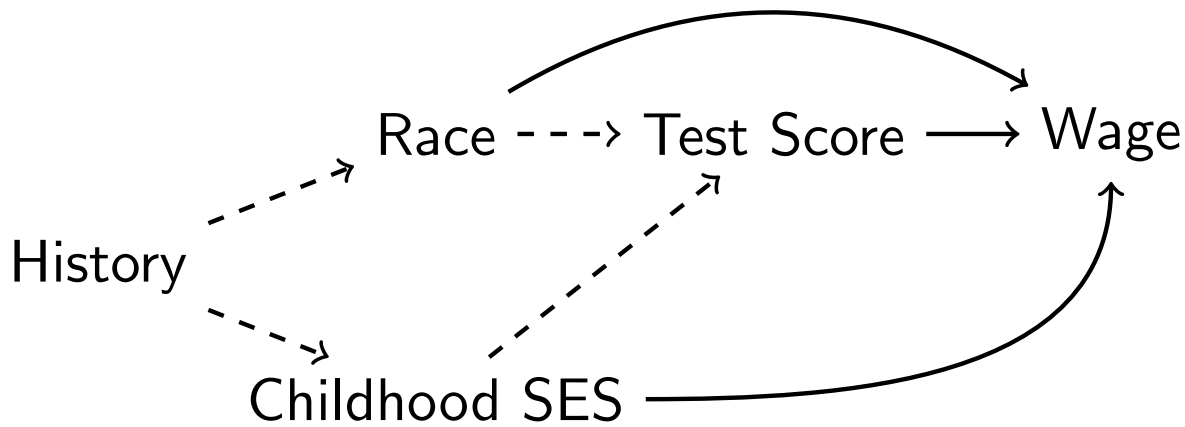
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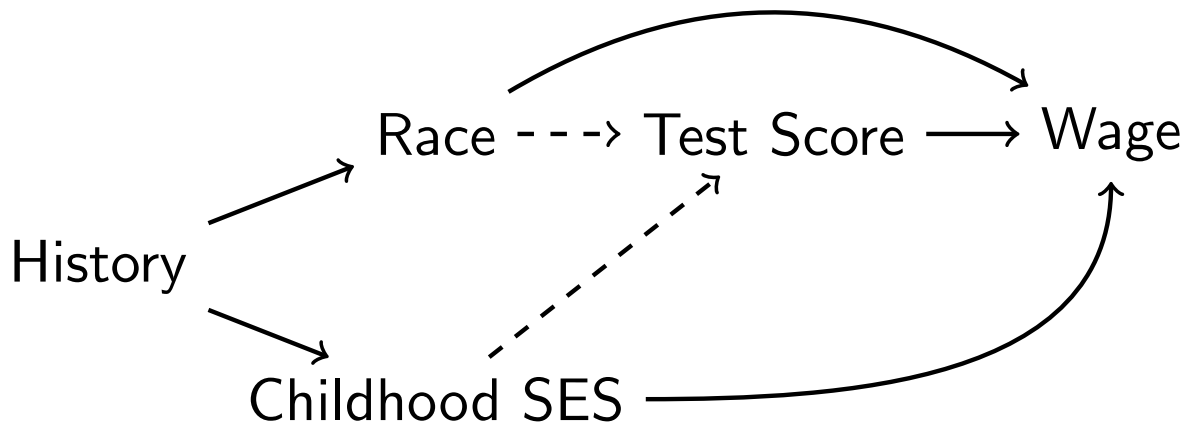
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Past work: Causal decomposition analysis

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Choice of intervention targets — Jackson & Vanderweele 2018

Equity: What should we equalize? — Jackson 2021

Past work: Causal decomposition analysis

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Choice of intervention targets — Jackson & Vanderweele 2018

Equity: What should we equalize? — Jackson 2021

Systems may adapt to maintain inequity — Jackson & Arah 2020

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- Impact: Others using these ideas

Define an intervention

Using the Chetty et al. 2017 example,

What gap in respondent incomes would remain
across categories of parent income
if we intervened to send people to selective colleges?

Interpret with respect to a **target trial** (Hernán & Robins 2016)

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1. Sample $\mathcal{S}$ from the population

Interpret with respect to a **target trial** (Hernán & Robins 2016)

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2. Assign treatment $T = 1$ to $\mathcal{S}$

Interpret with respect to a **target trial** (Hernán & Robins 2016)

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity
   across categories X

Interpret with respect to a **target trial** (Hernán & Robins 2016)

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Goal: Expected result over
hypothetical samples $\mathcal{S}$

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

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Global intervention

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Global intervention

1. Take the entire population $\mathcal{P}$

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Global intervention

1. Take the entire population $\mathcal{P}$
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Goal: Expected result over hypothetical samples $\mathcal{S}$

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

Goal: Result of this procedure

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Difficulty: Causal inference

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

Goal: Result of this procedure

Difficulty: Causal inference

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Difficulty: Causal inference

Global intervention

1. Take the entire population $\mathcal{P}$
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Goal: Result of this procedure

Difficulty: Causal inference
Equilibrium dynamics

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Equilibrium dynamics

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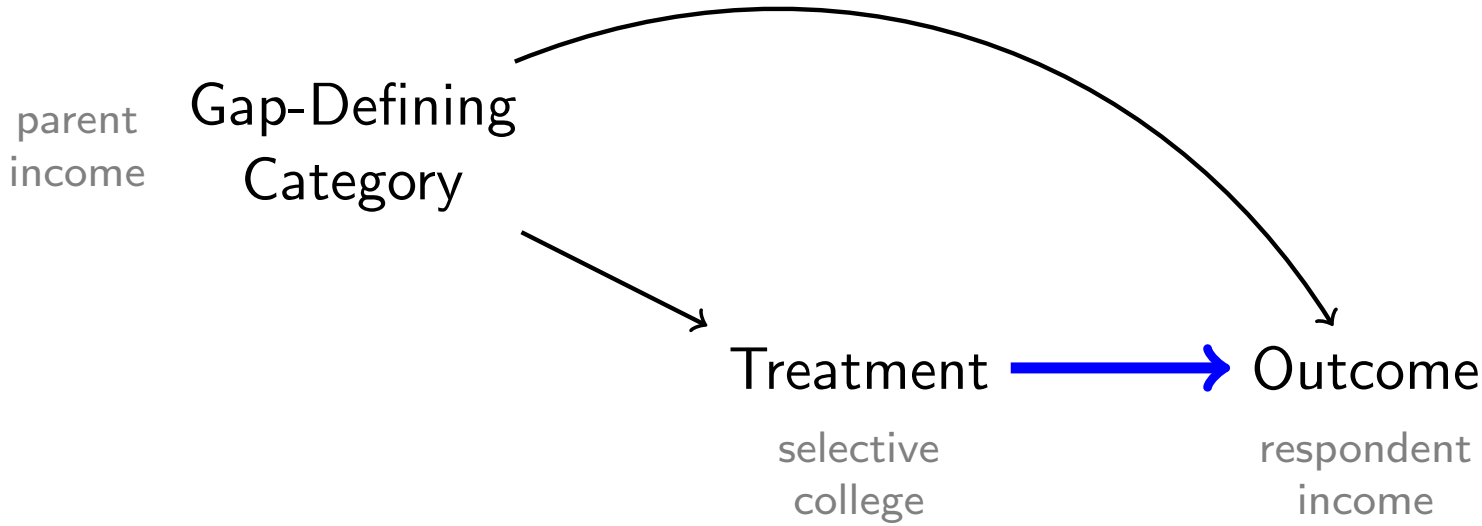
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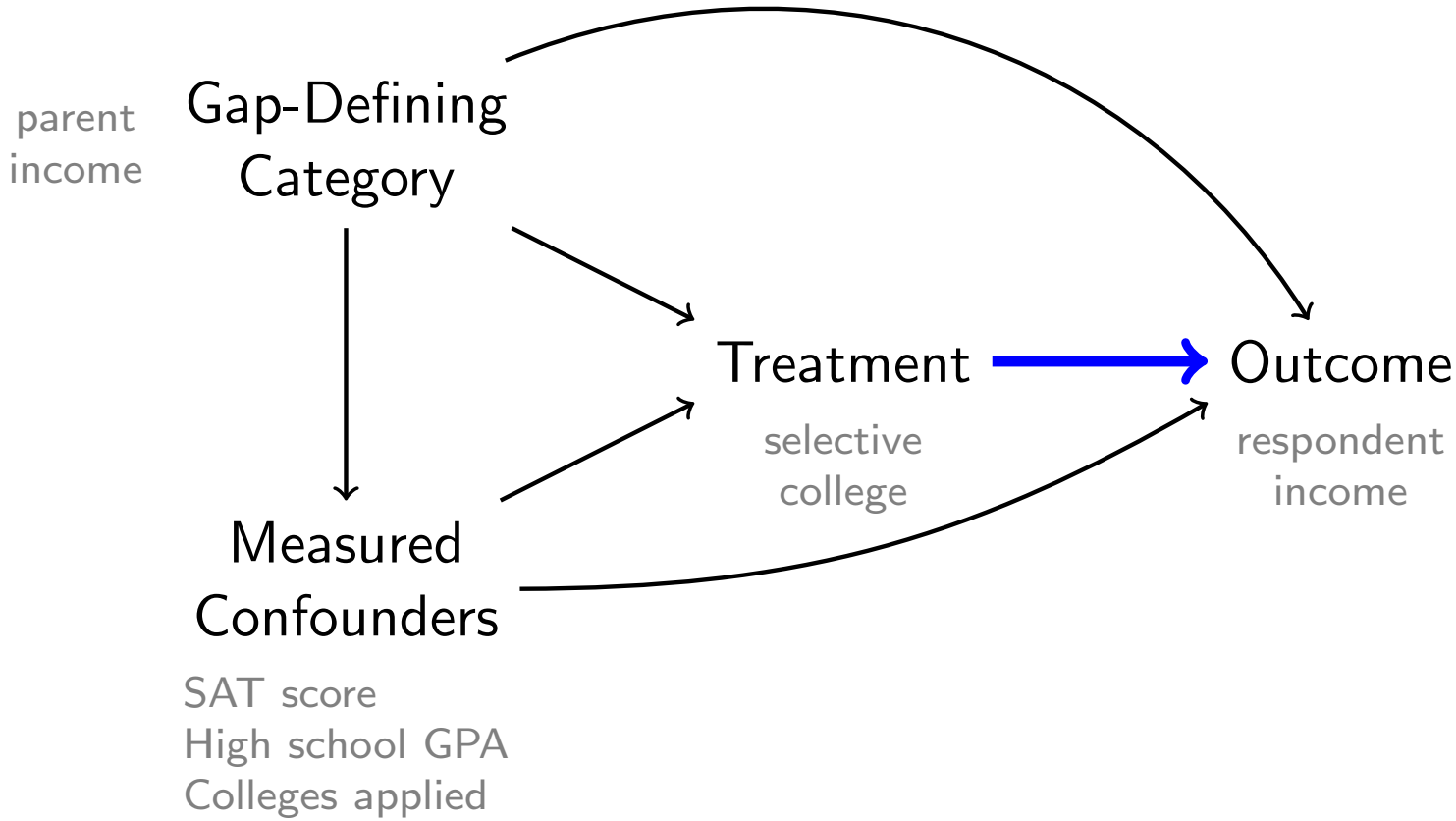
The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

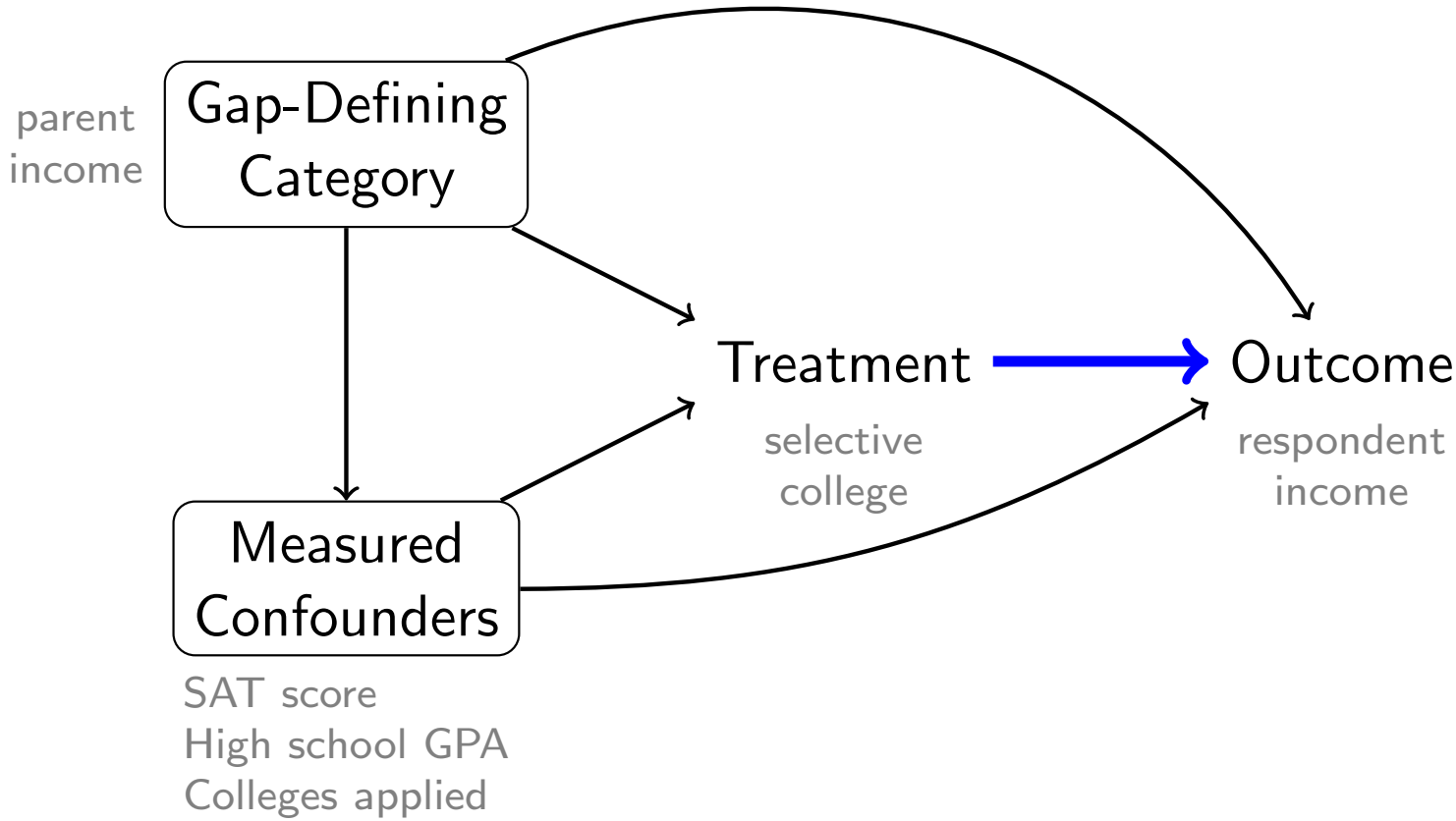
- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention?
- **Identify the estimand**
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- Produce software
- Impact: Others using these ideas


parent
income
Gap-Defining
Category
Treatment
selective
college
Outcome
respondent
income

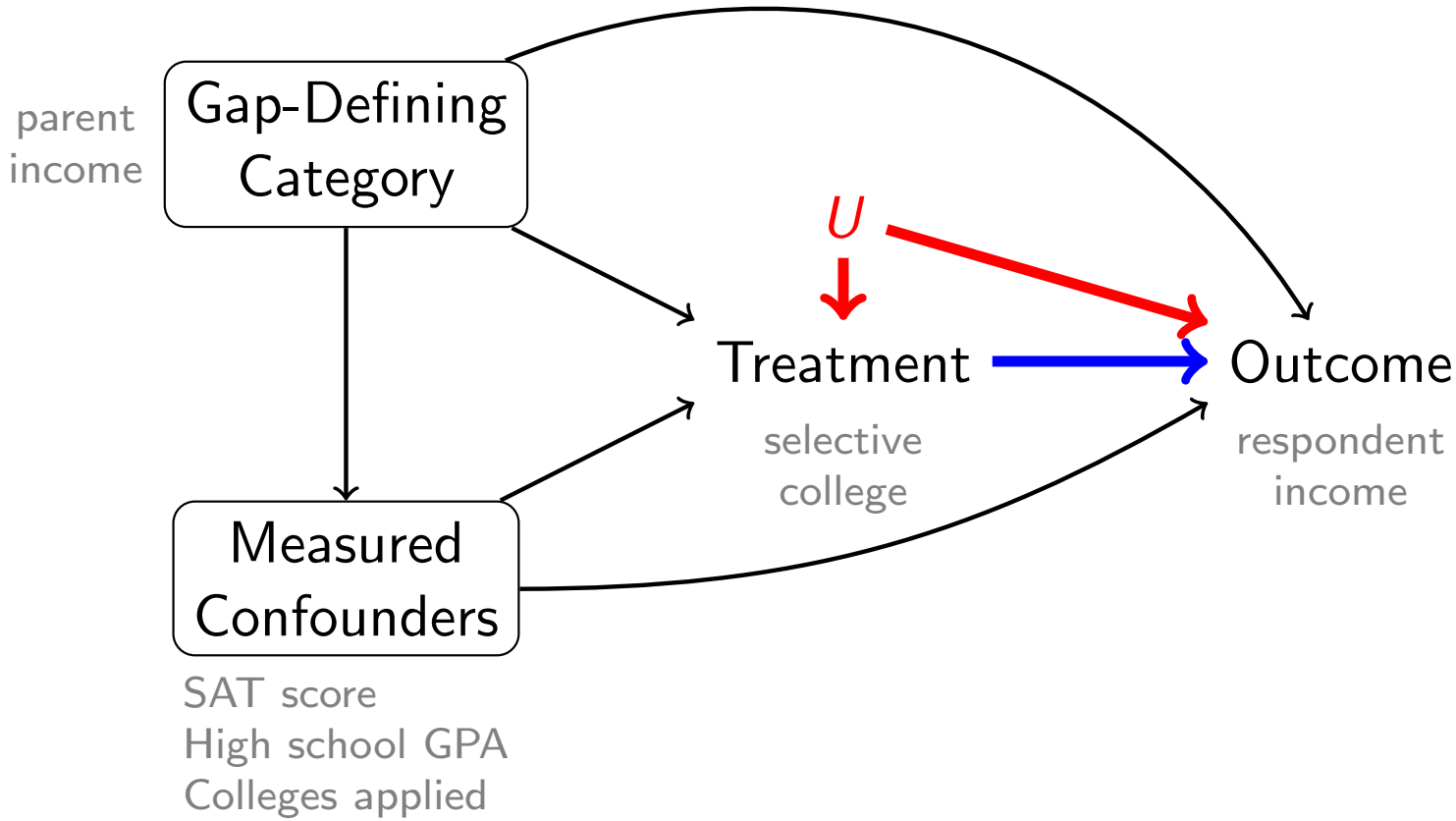
Pearl 2009


parent income
Gap-Defining Category
Treatment
selective college
Outcome
respondent income
Measured Confounders
SAT score
High school GPA
Colleges applied

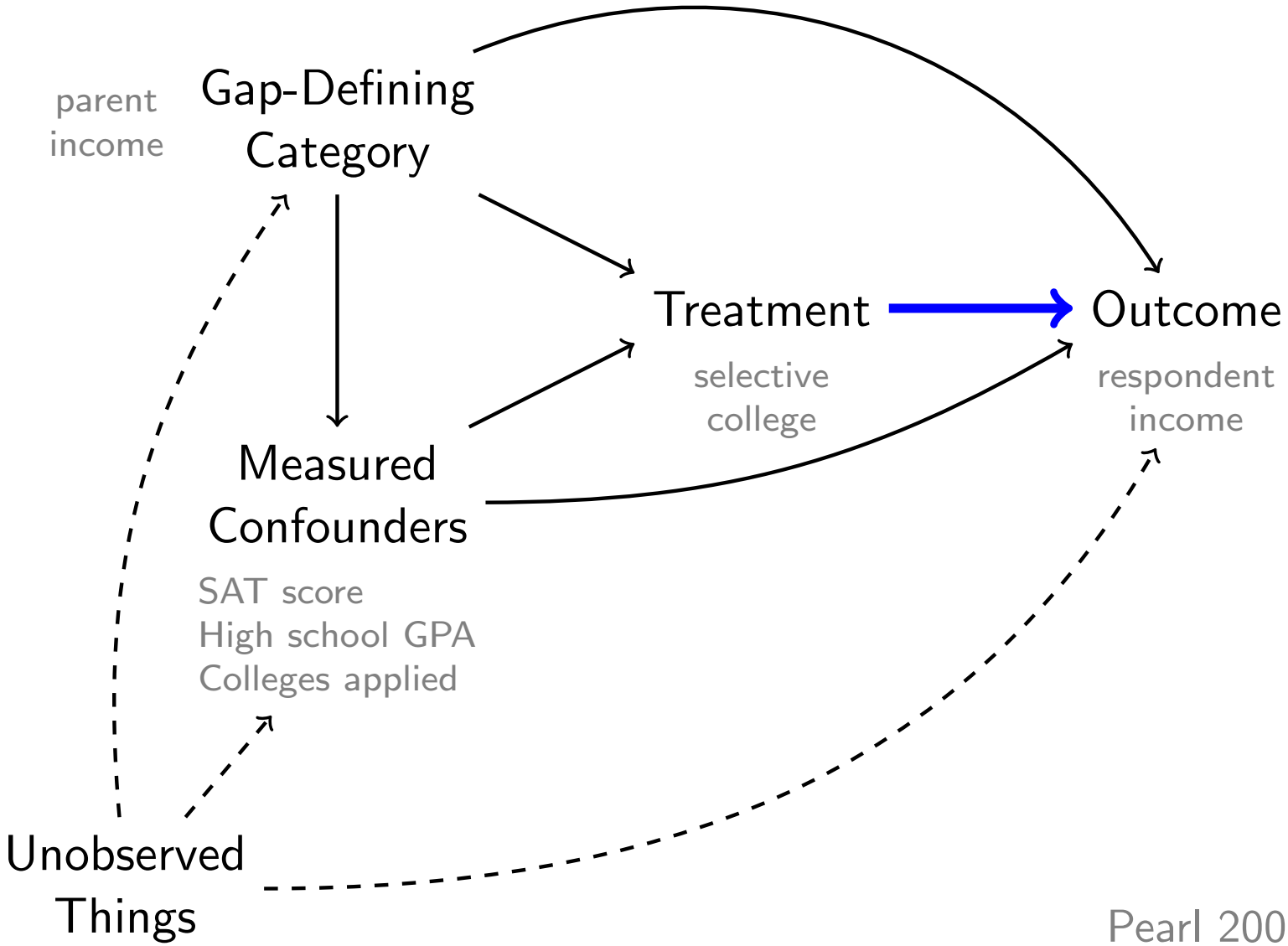
Pearl 2009


parent income
Gap-Defining Category
Treatment
selective college
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Measured Confounders
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High school GPA
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Pearl 2009


parent
income
Gap-Defining
Category
U
Treatment
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Pearl 2009


parent income
Gap-Defining Category
Treatment
selective college
Outcome
respondent income
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High school GPA
Colleges applied
Unobserved Things

Pearl 2009

The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

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- Impact: Others using these ideas

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
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Learn a prediction function

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	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

Predict the whole table

		Outcome under treatment	Outcome under control
People in category 1	Person 1	$\hat{Y}_1(1)$	$\hat{Y}_1(0)$
	Person 2	$\hat{Y}_2(1)$	$\hat{Y}_2(0)$
	Person 3	$\hat{Y}_3(1)$	$\hat{Y}_3(0)$
People in category 2	Person 4	$\hat{Y}_4(1)$	$\hat{Y}_4(0)$
	Person 5	$\hat{Y}_5(1)$	$\hat{Y}_5(0)$
	Person 6	$\hat{Y}_6(1)$	$\hat{Y}_6(0)$

Robins 1986
Hahn 1998

Learn a prediction function

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Problem: Optimization for the wrong task

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Problem: Optimization for the wrong task

Prediction error over
observed
cases

Learn a prediction function

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
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Predict the whole table

		Outcome under treatment	Outcome under control
People in category 1	Person 1	$\hat{Y}_1(1)$	$\hat{Y}_1(0)$
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Problem: Optimization for the wrong task

Prediction error over **observed** cases

vs

Prediction error over **all** cases

Solution: Reweight errors to approximate the correct task

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		Prediction under treatment
People in category 1	Person 1	$\hat{Y}_1(1)$
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		Prediction under treatment	Outcome under treatment
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	Person 5	$\hat{Y}_5(1)$	Y_5
	Person 6	$\hat{Y}_6(1)$	?

Solution: Reweight errors to approximate the correct task

		Prediction under treatment	Outcome under treatment	Error
People in category 1	Person 1	$\hat{Y}_1(1)$	?	?
	Person 2	$\hat{Y}_2(1)$	Y_2	$\hat{Y}_2(1) - Y_2$
	Person 3	$\hat{Y}_3(1)$	Y_3	$\hat{Y}_3(1) - Y_3$
People in category 2	Person 4	$\hat{Y}_4(1)$	?	?
	Person 5	$\hat{Y}_5(1)$	Y_5	$\hat{Y}_5(1) - Y_5$
	Person 6	$\hat{Y}_6(1)$	?	?

Solution: Reweight errors to approximate the correct task

		Prediction under treatment	Outcome under treatment	Error	Weight on error
People in category 1	Person 1	$\hat{Y}_1(1)$	?	?	
	Person 2	$\hat{Y}_2(1)$	Y_2	$\hat{Y}_2(1) - Y_2$	3 / 2
	Person 3	$\hat{Y}_3(1)$	Y_3	$\hat{Y}_3(1) - Y_3$	3 / 2
People in category 2	Person 4	$\hat{Y}_4(1)$	?	?	
	Person 5	$\hat{Y}_5(1)$	Y_5	$\hat{Y}_5(1) - Y_5$	3
	Person 6	$\hat{Y}_6(1)$	?	?	

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean$(\hat{Y}_i - Y_i)$ with
inverse probability of treatment weights

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People in category 1	Person 1	$\hat{Y}_1(1)$	?	?	
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People in category 2	Person 4	$\hat{Y}_4(1)$	?	?	
	Person 5	$\hat{Y}_5(1)$	Y_5	$\hat{Y}_5(1) - Y_5$	3
	Person 6	$\hat{Y}_6(1)$	?	?	

Solution: Reweight errors to approximate the correct task

Estimated bias:	Mean$(\hat{Y}_i - Y_i)$ with inverse probability of treatment weights
New Estimate:	(Original Estimate) $-$ (Estimated Bias)

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
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New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

Robins, Rotnitzky, & Zhao 1994
Bang & Robins 2005

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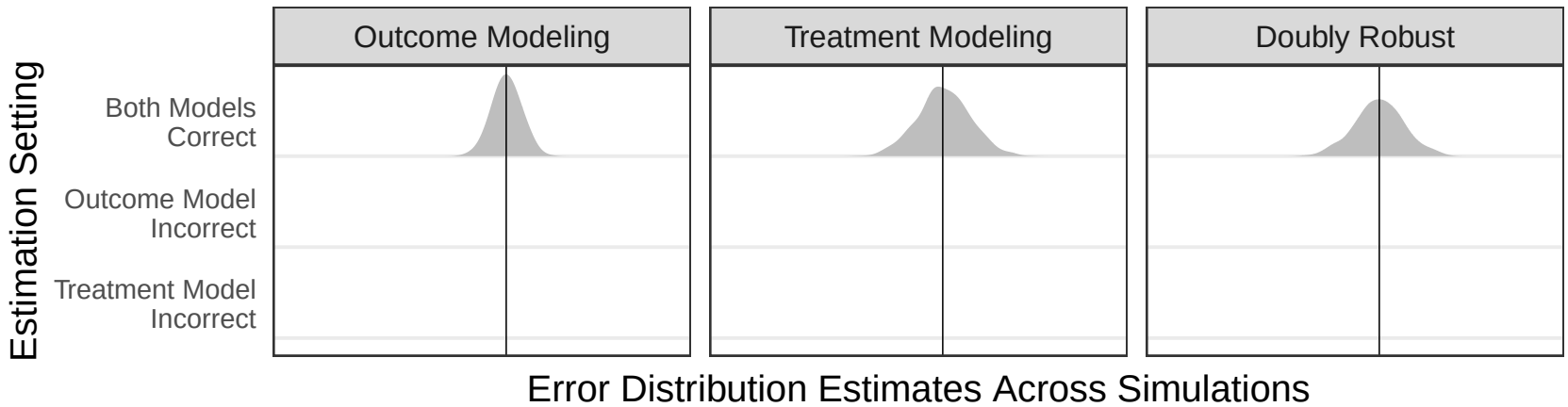
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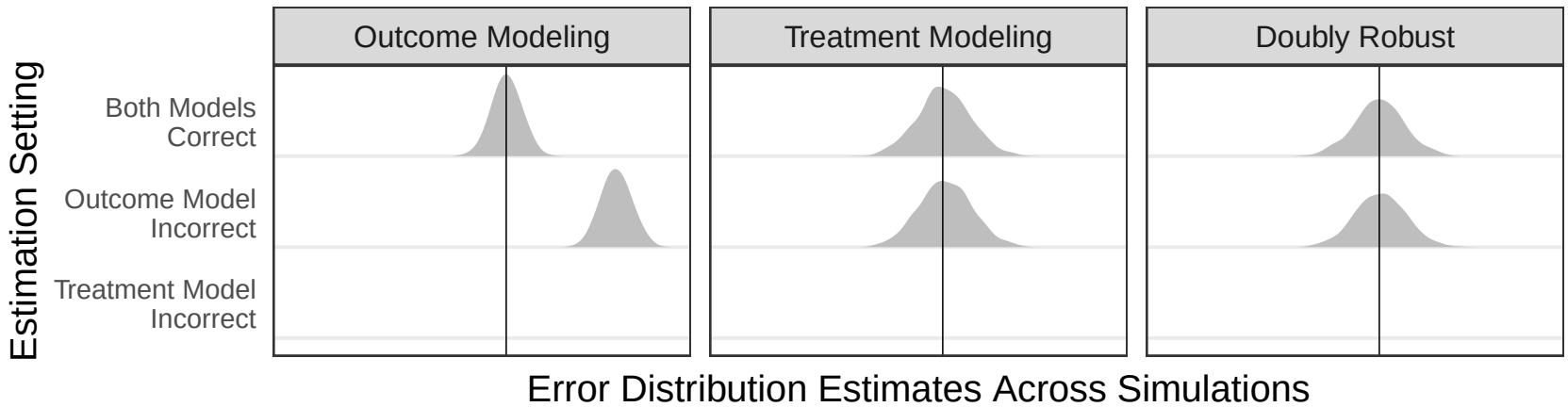
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New Estimate: (Original Estimate) – (Estimated Bias)

Doubly Robust Estimation



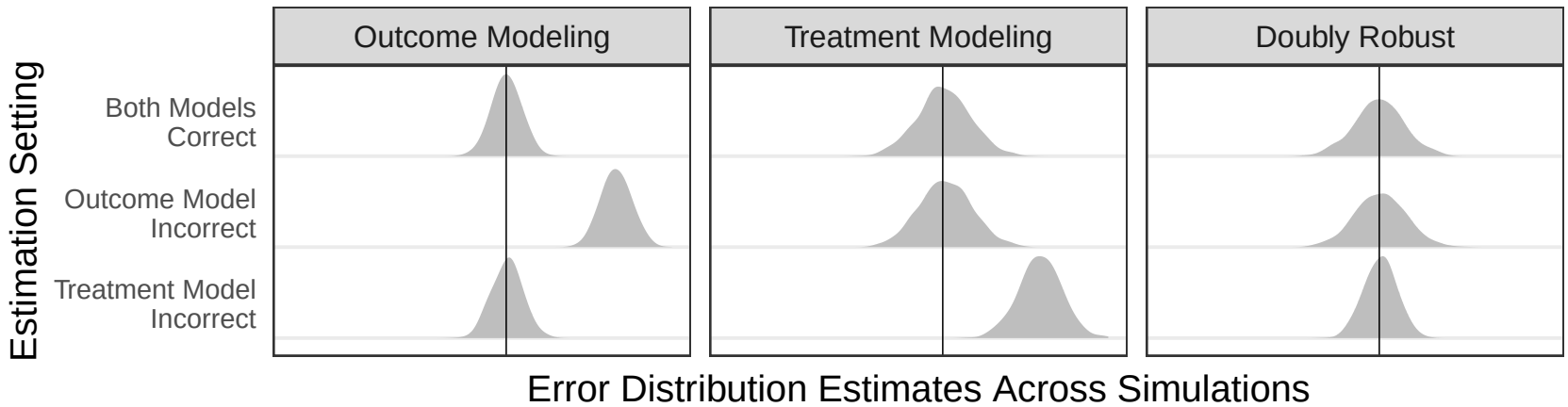
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Doubly Robust Estimation

Even better:

Chernozhukov et al. 2018
Bickel 1982

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
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New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

Even better: — Learn $\hat{Y}_i$ in sample A

— Estimate bias in sample B

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Even better:
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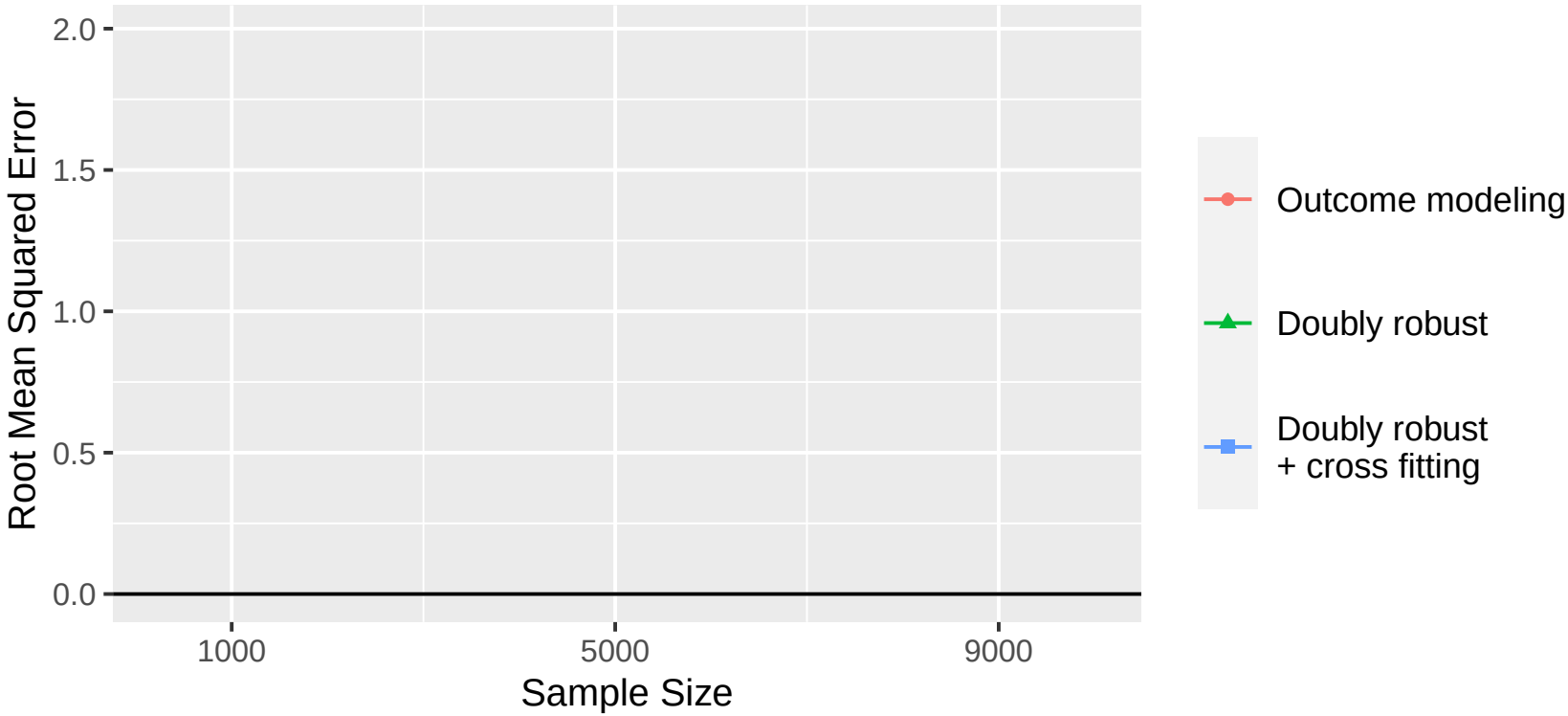
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Doubly Robust Estimation

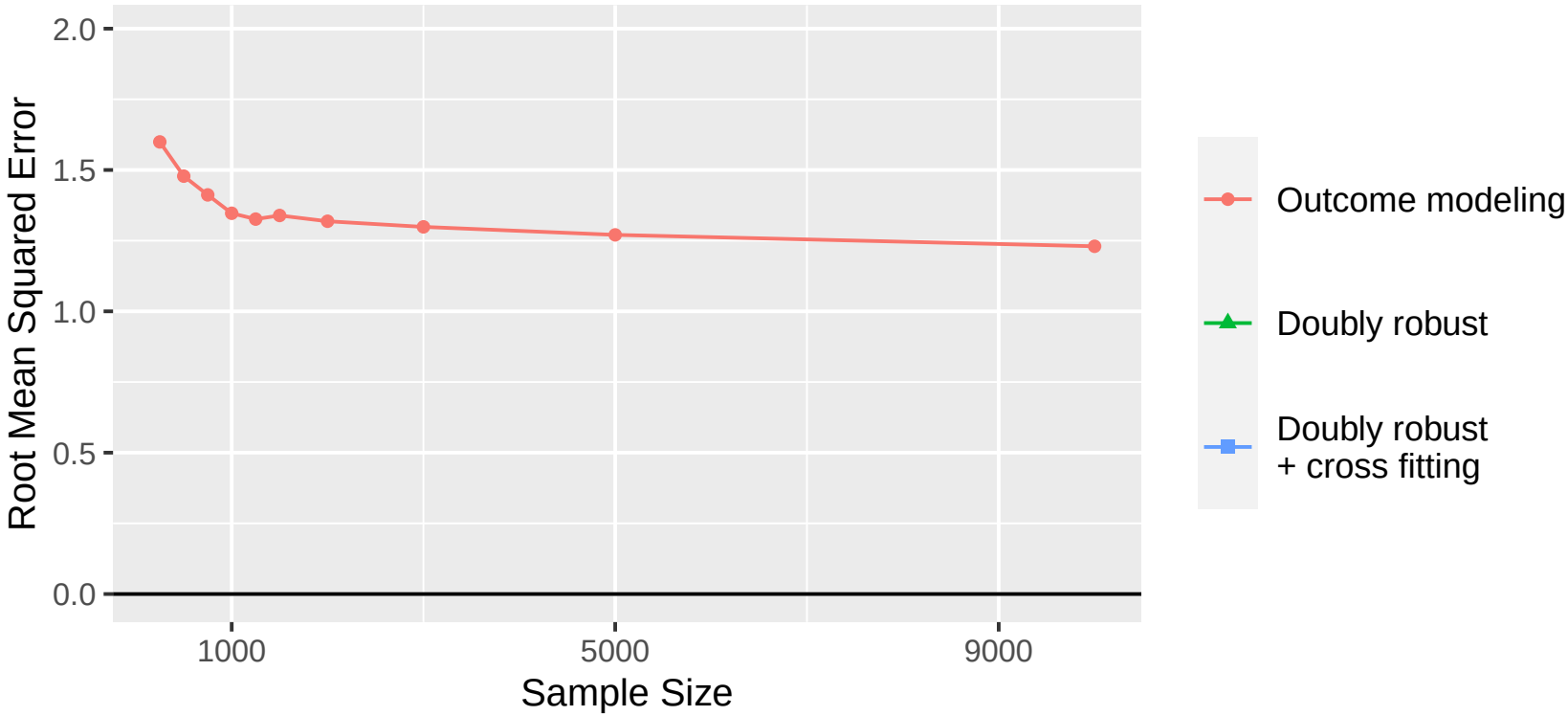
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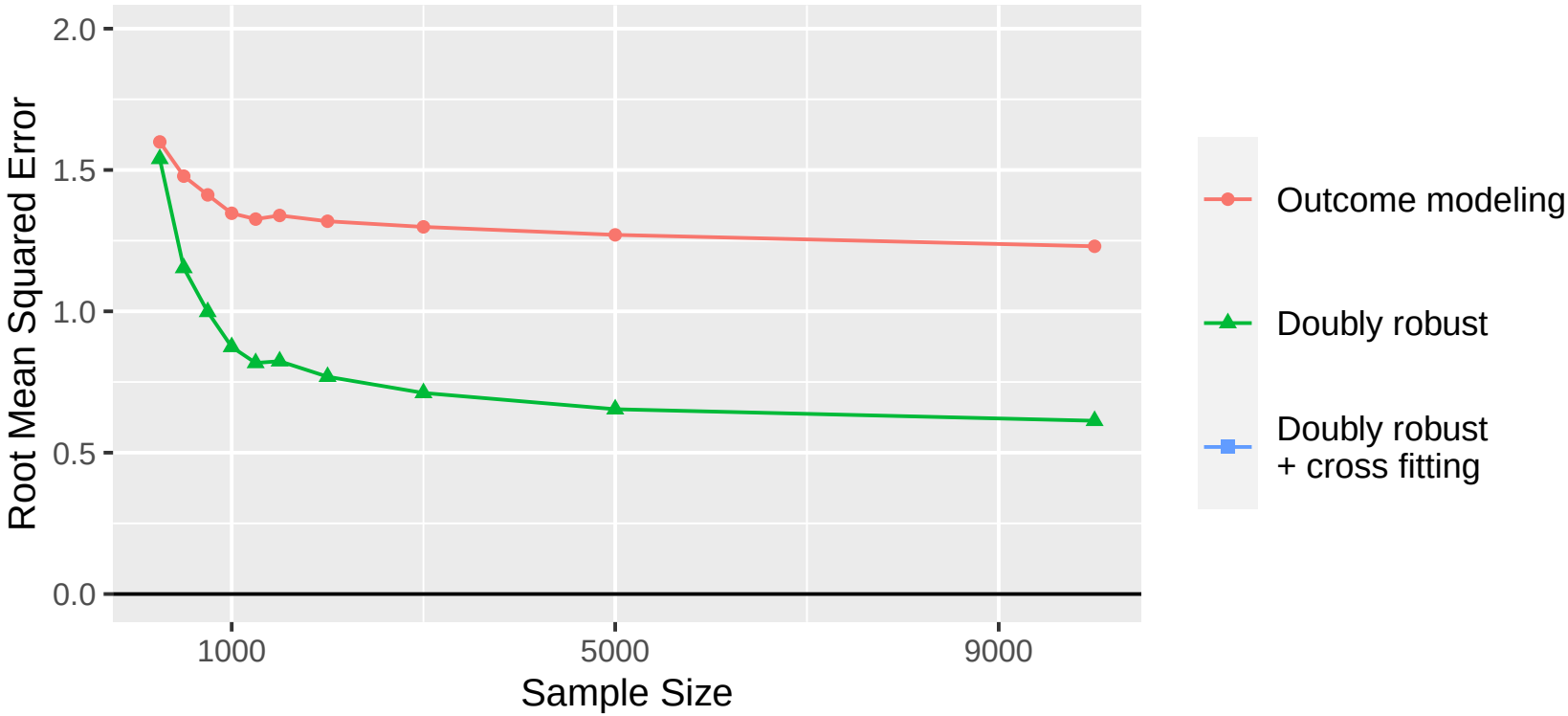
Solution: Reweight errors to approximate the correct task



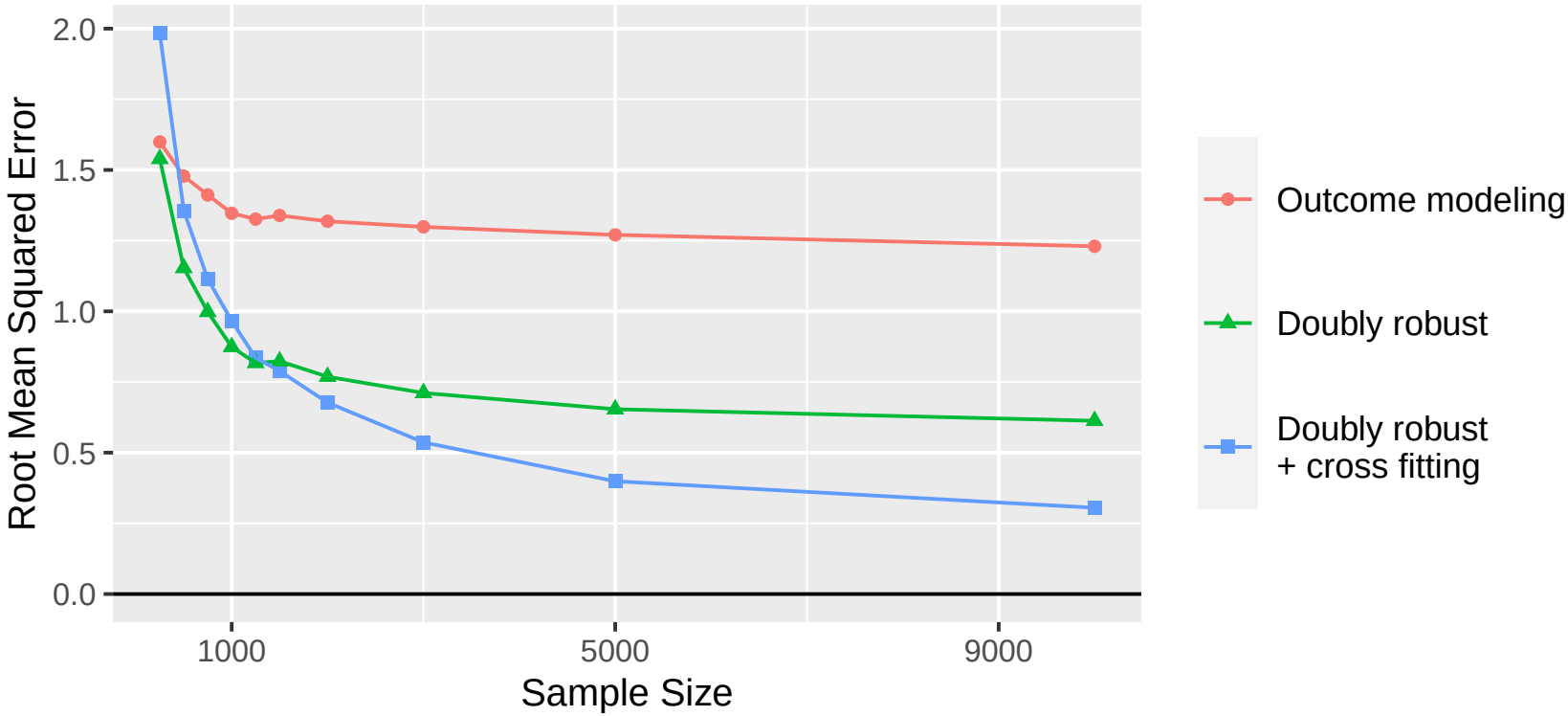
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Solution: Reweight errors to approximate the correct task



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Estimated bias: Mean($\hat{Y}_i - Y_i$) with
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Double Machine Learning

Chernozhukov et al. 2018
Bickel 1982

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Double Machine Learning



The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

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gapclosing

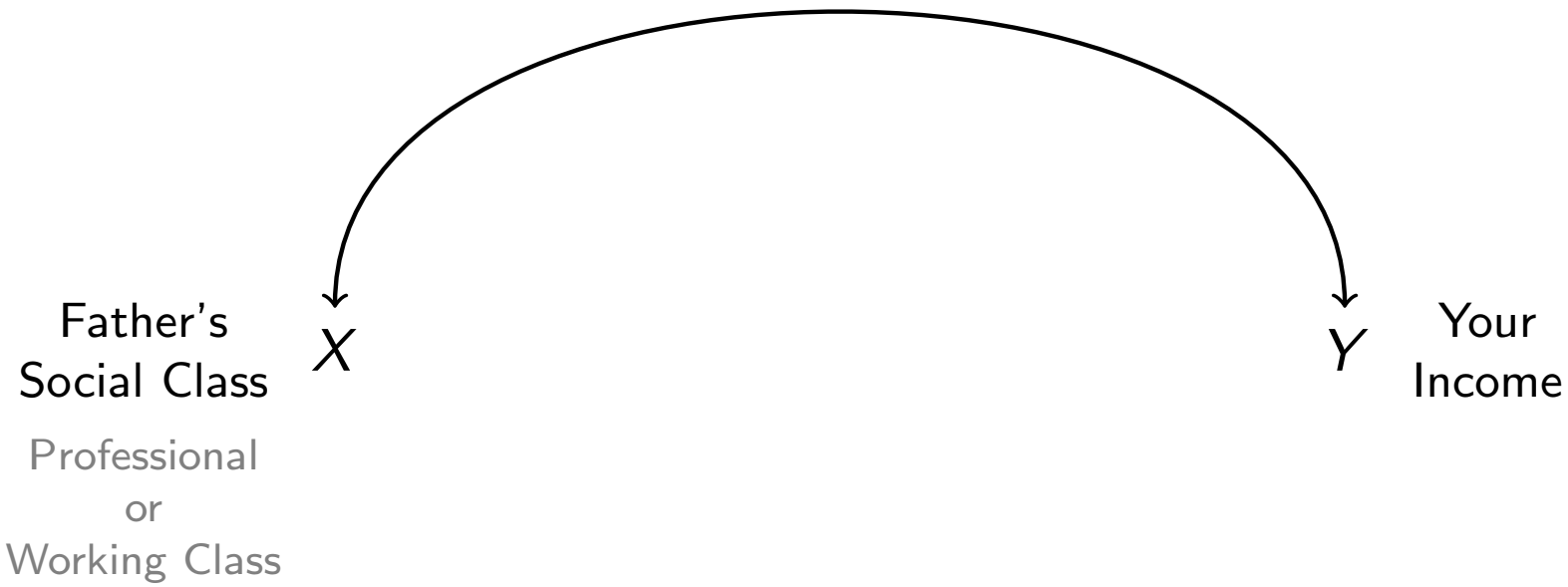
An R package to estimate gap closing estimands. Install this package with the command `devtools::install_github("ilundberg/gapclosing")`.

```
estimate <- gapclosing(
  data = simulated_data,
  outcome_formula = formula(outcome ~ category + confounder),
  treatment_formula = formula(treatment ~ category + confounder),
  category_name = "category",
  counterfactual_assignments = 1,
  outcome_algorithm = "ranger",
  treatment_algorithm = "ranger",
  sample_split = "cross_fit",
  se = T
)
```

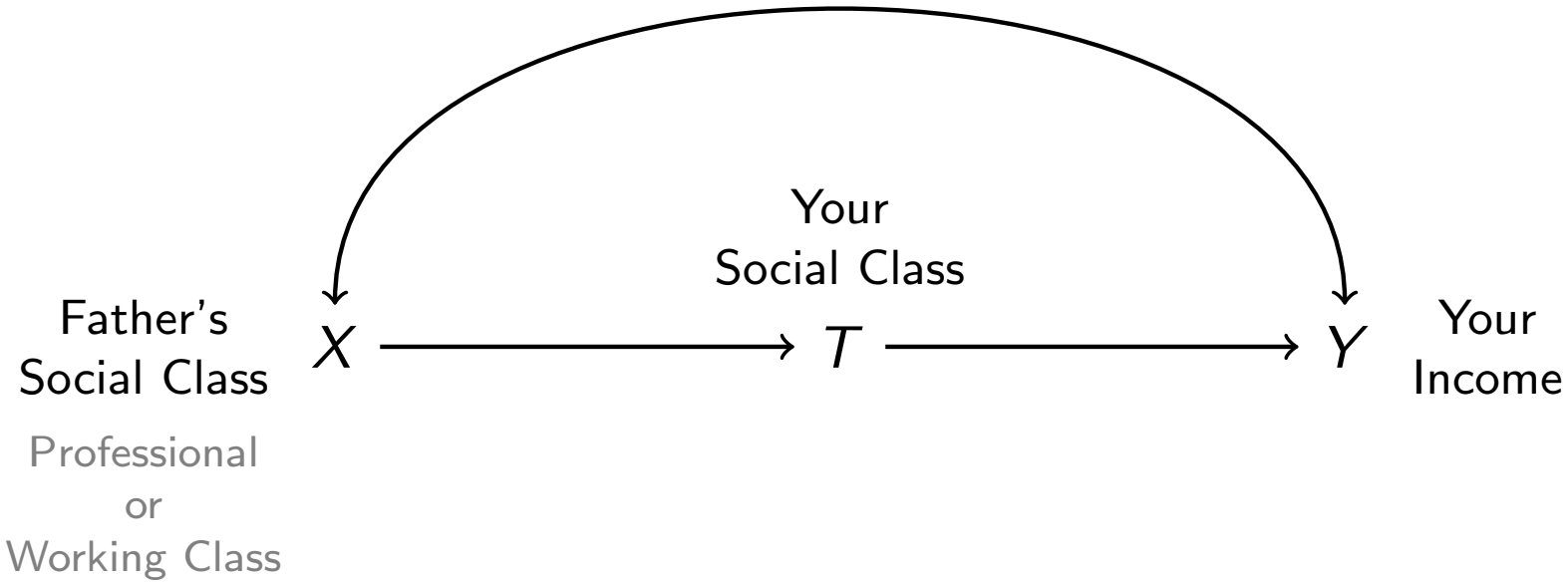
description	estimate	se	ci.min	ci.max
Factual gap	2.14	0.40	1.36	2.9
Counterfactual gap	0.67	0.44	-0.19	1.5

Empirical Examples

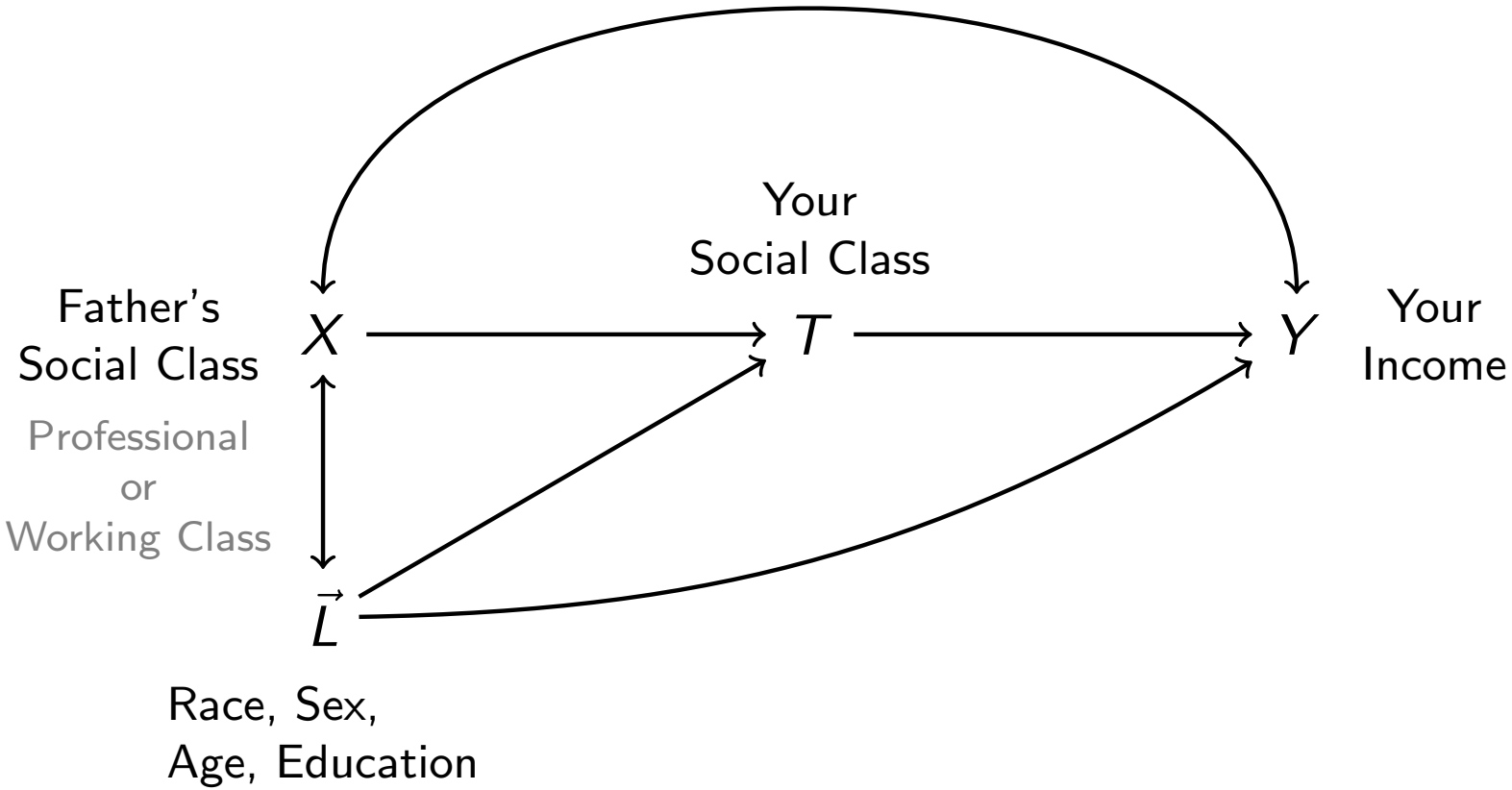
Empirical Example 1: Economic Mobility



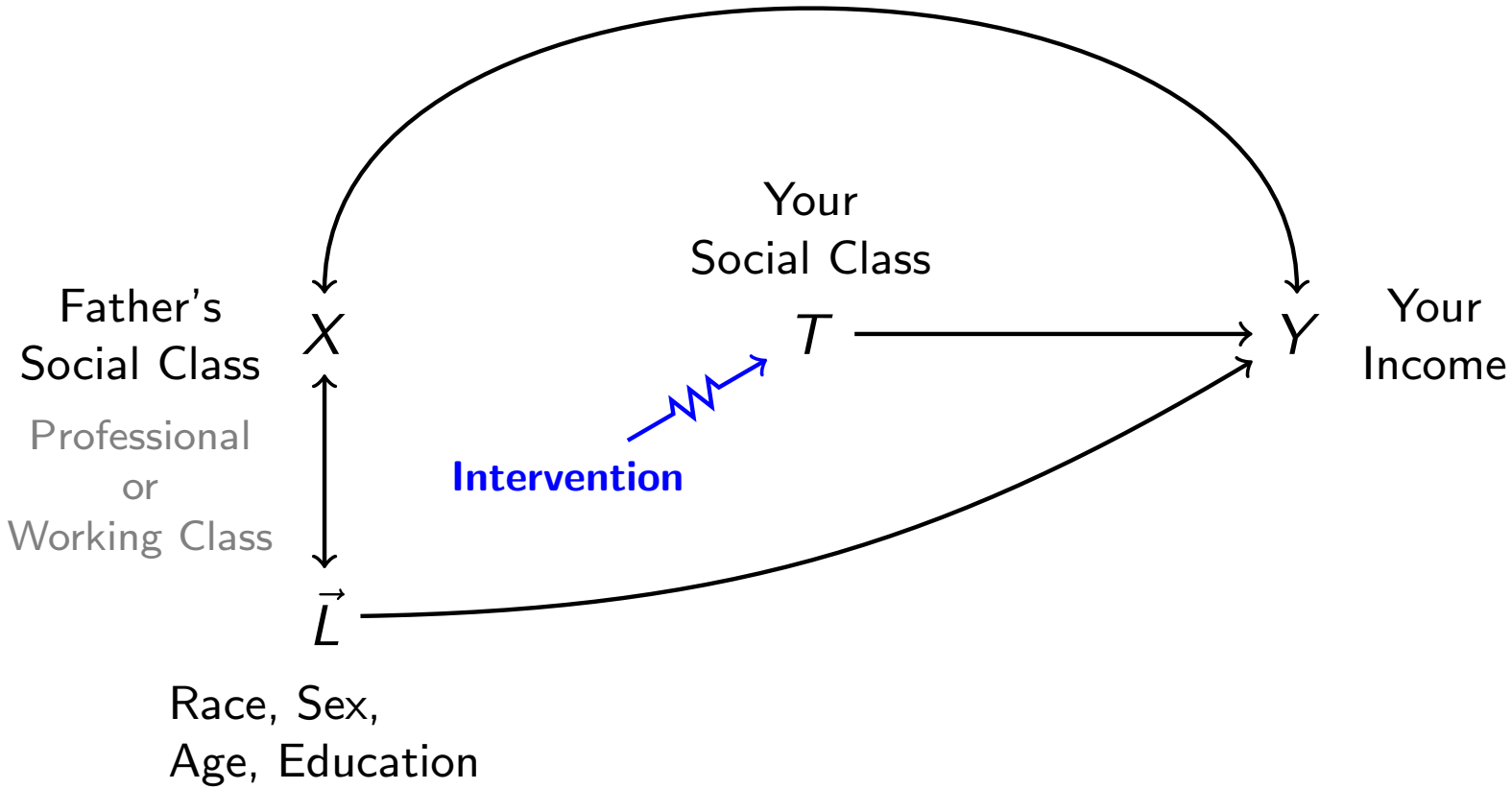
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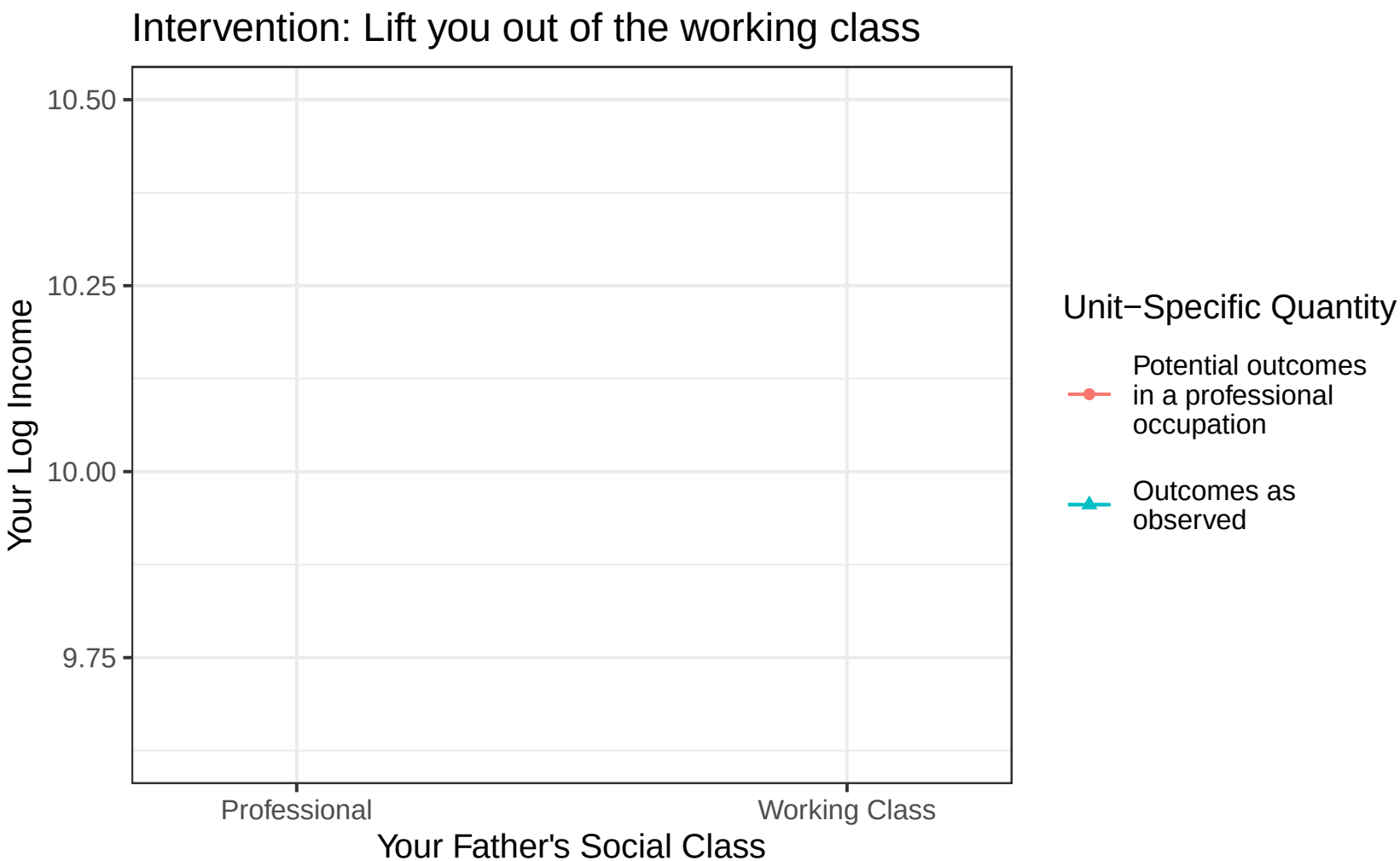
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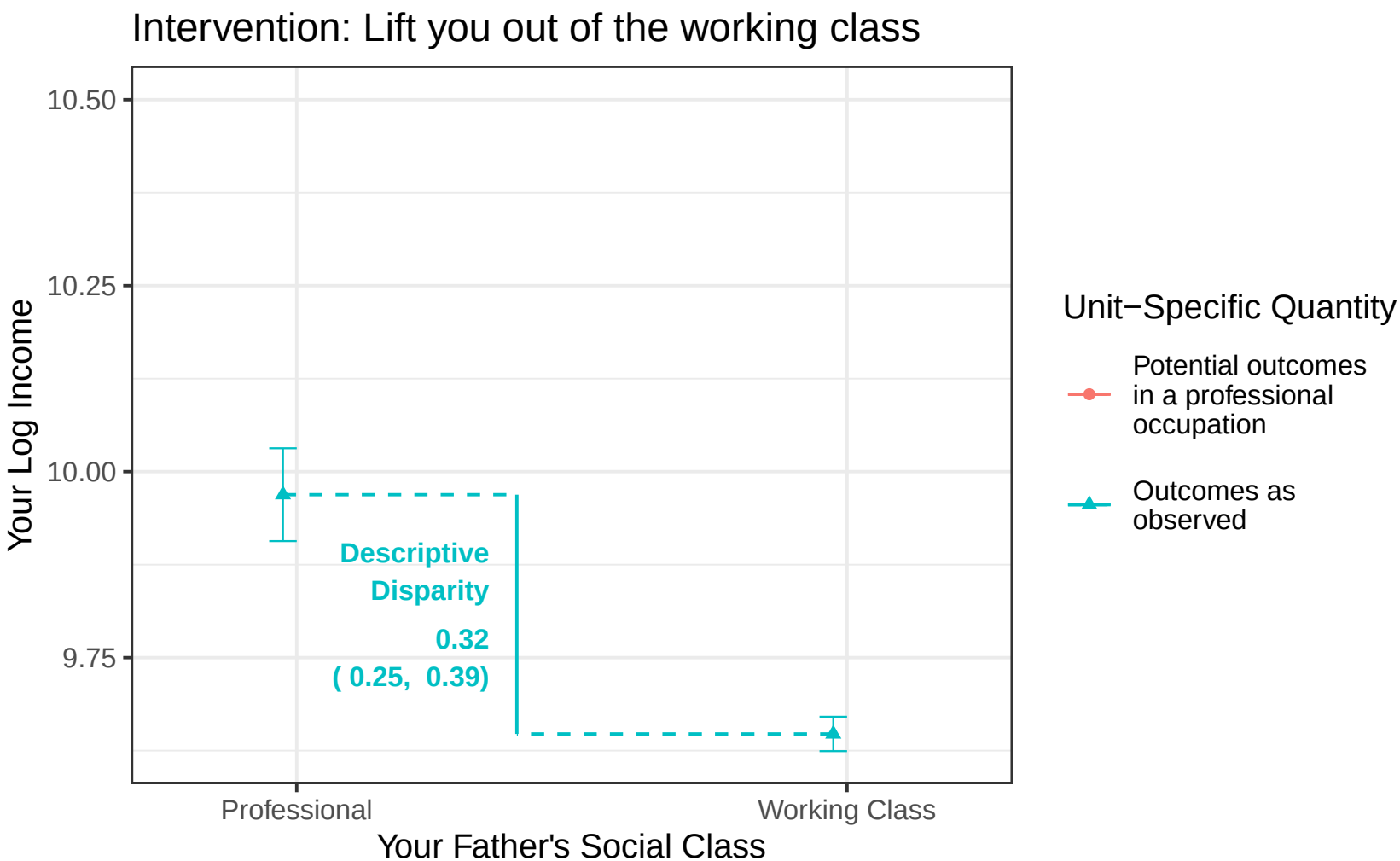
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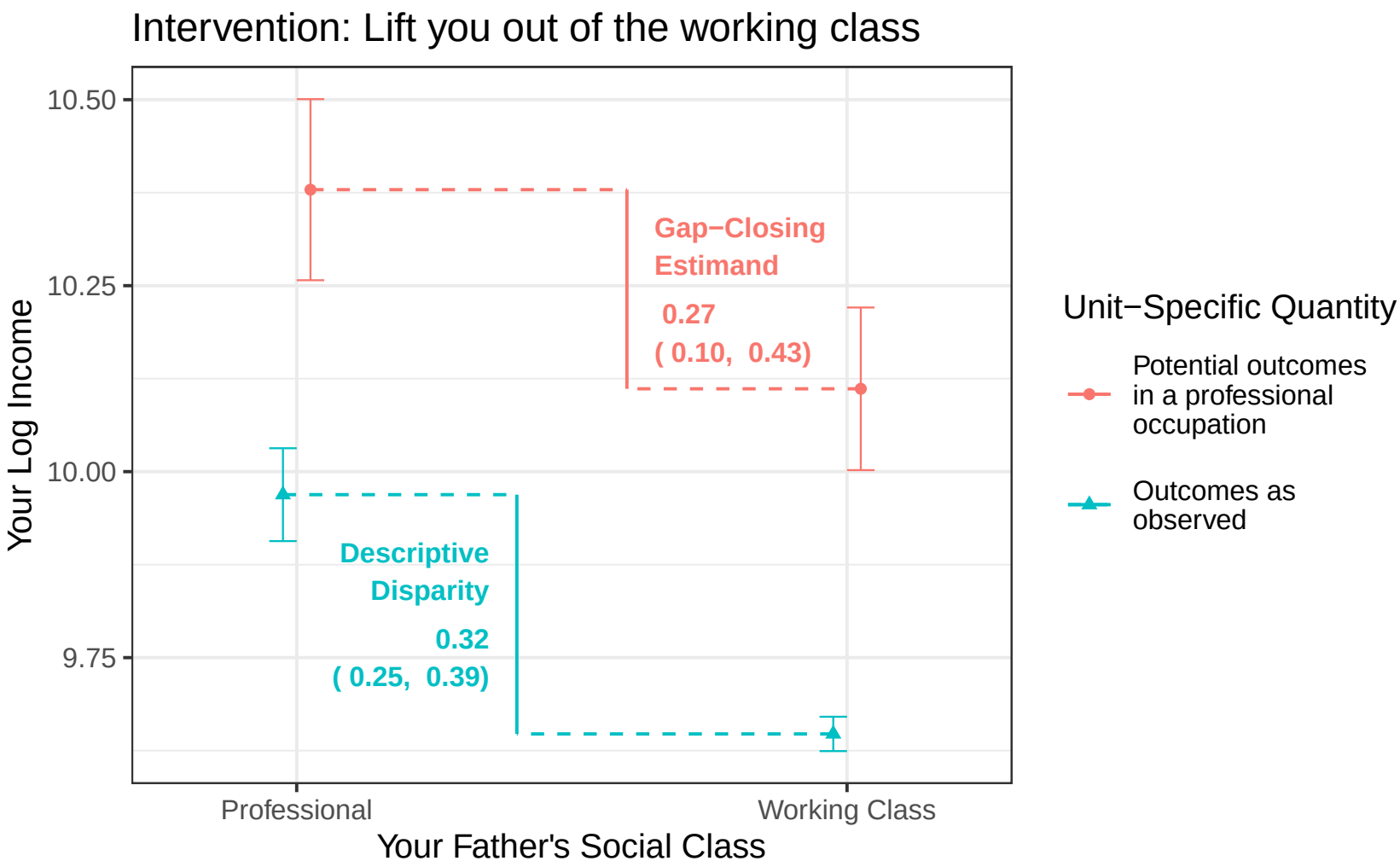
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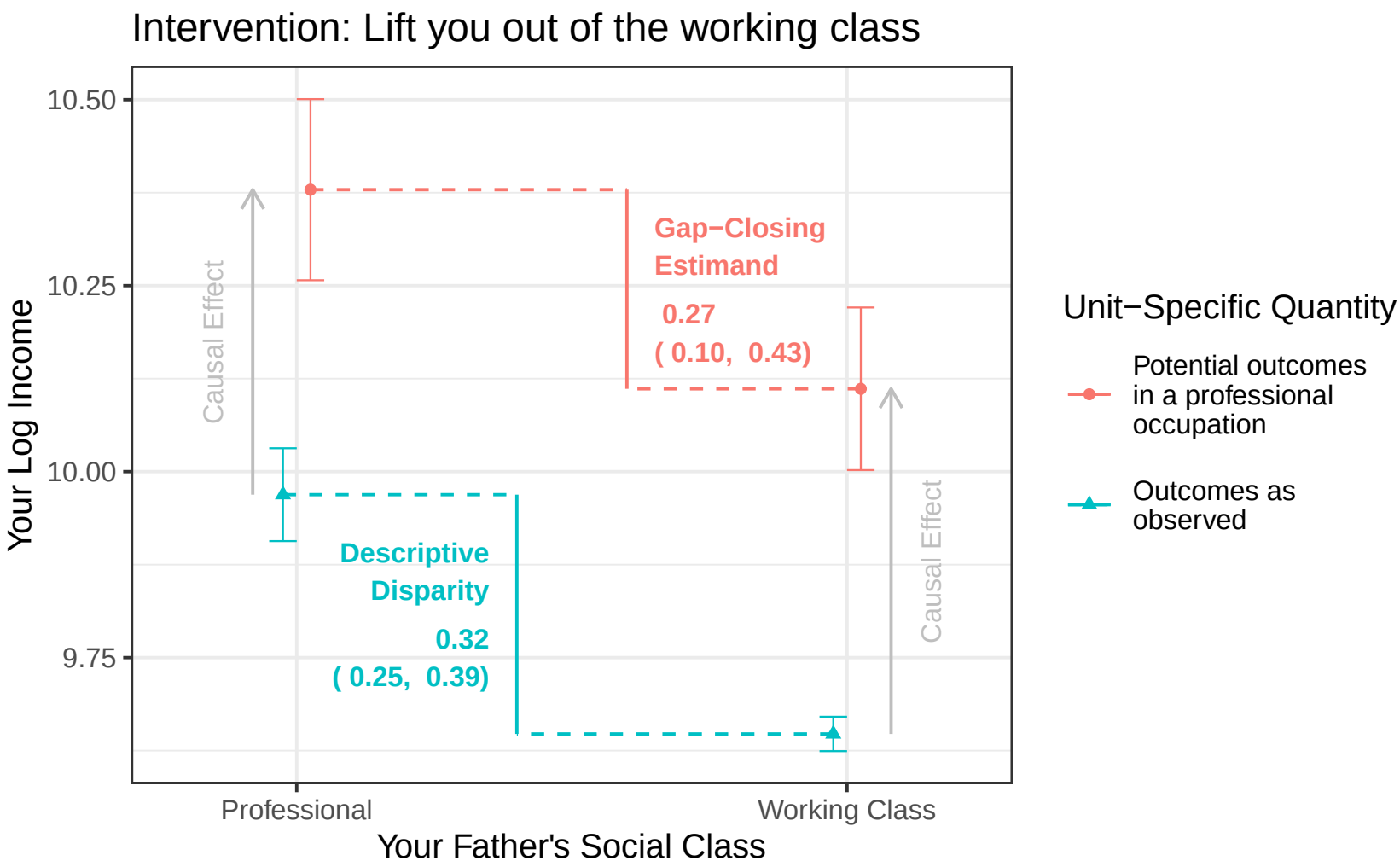
Empirical Example 1: Economic Mobility



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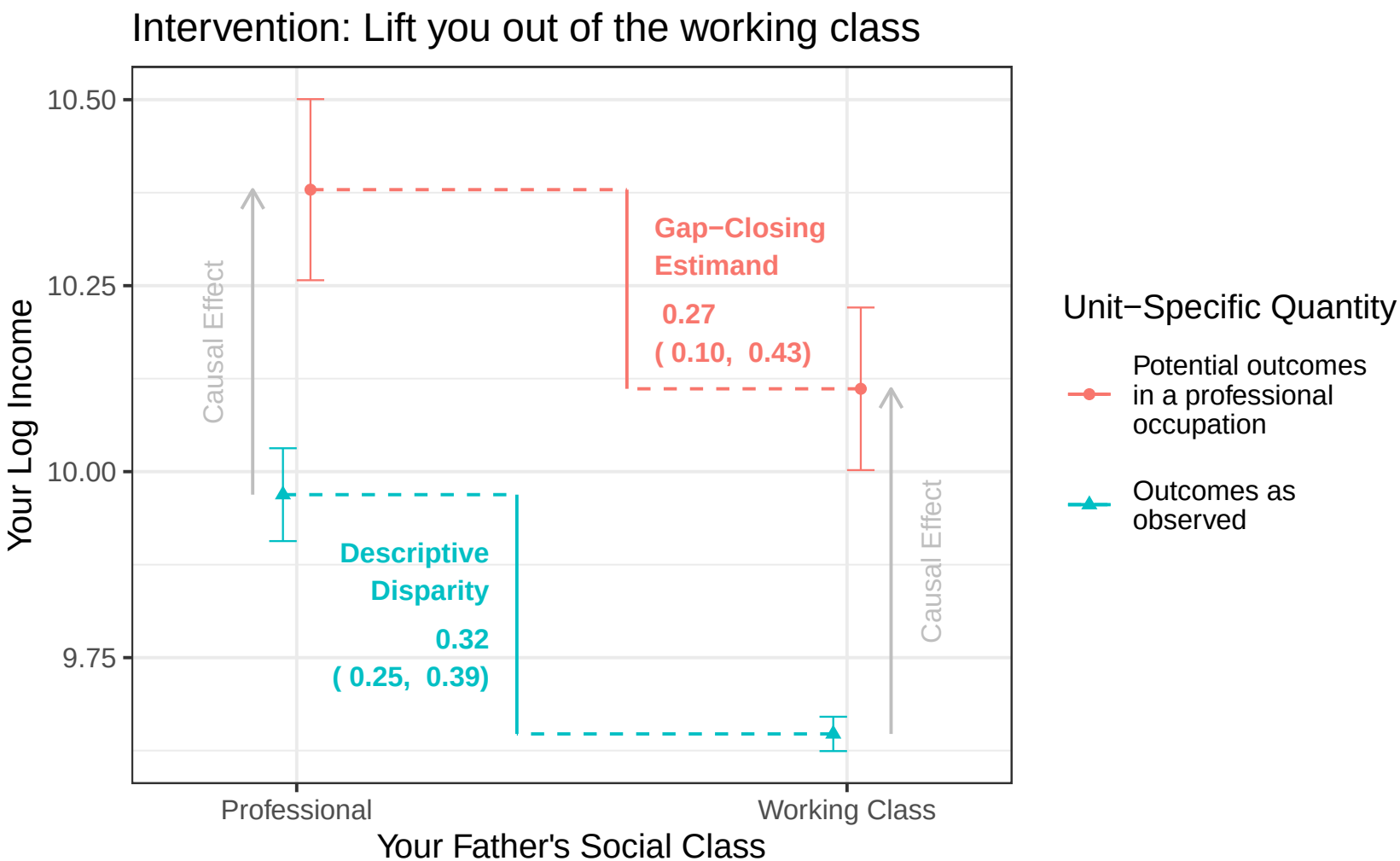


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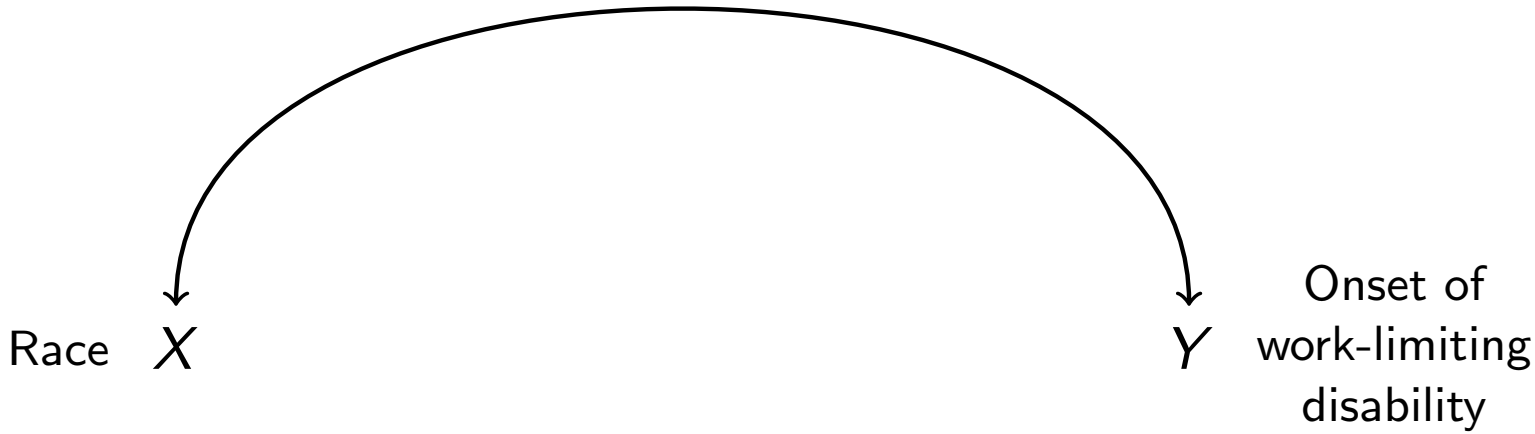


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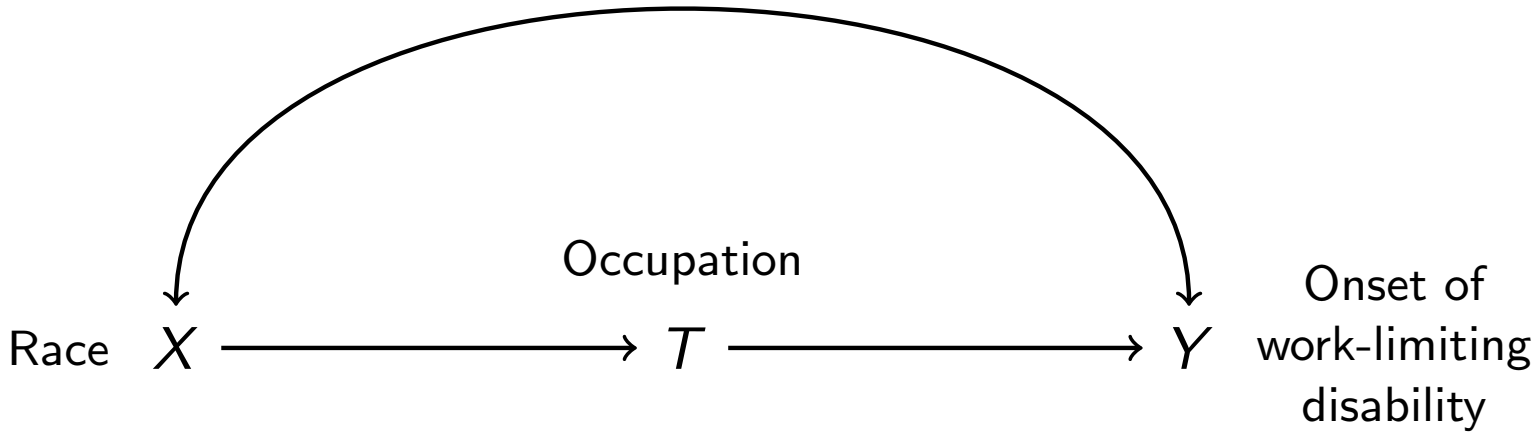
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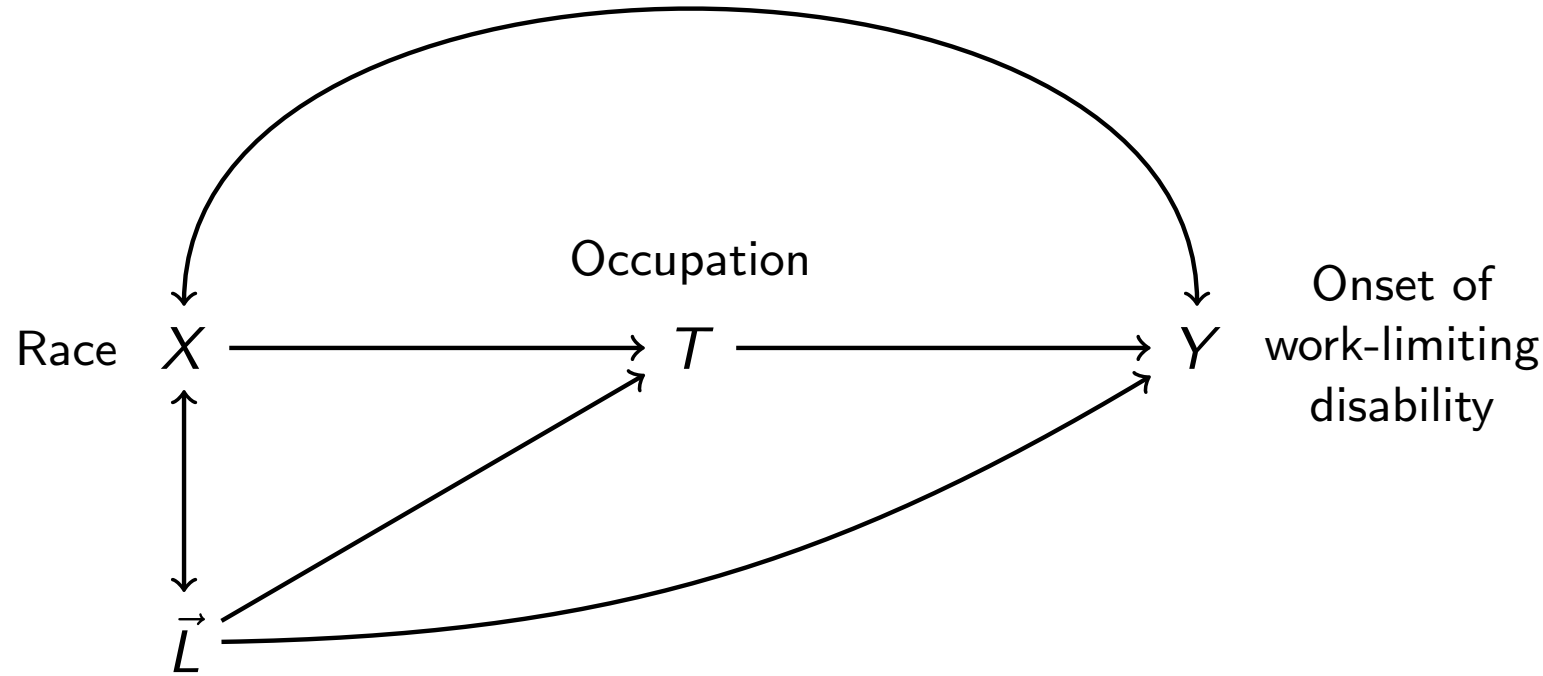
Empirical Example 2: Racial Disparities in Health



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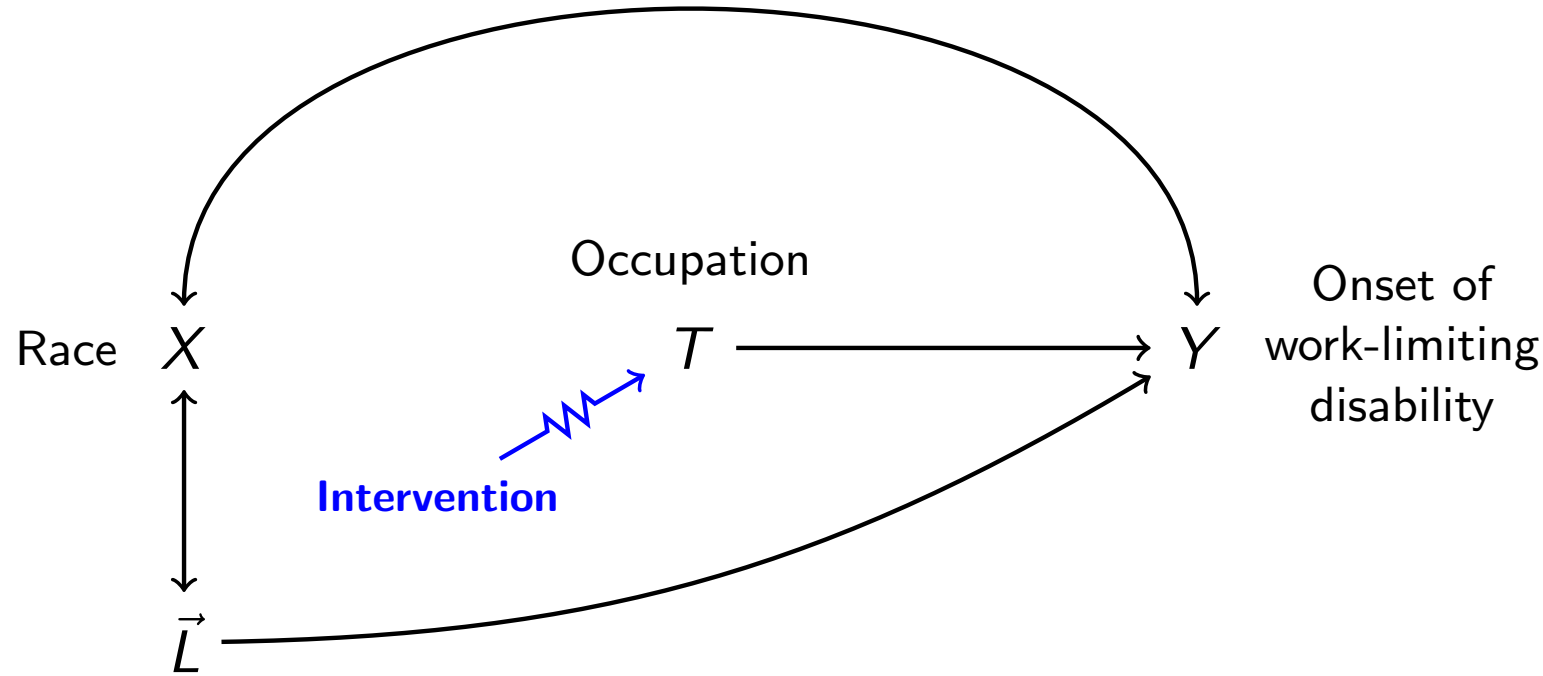


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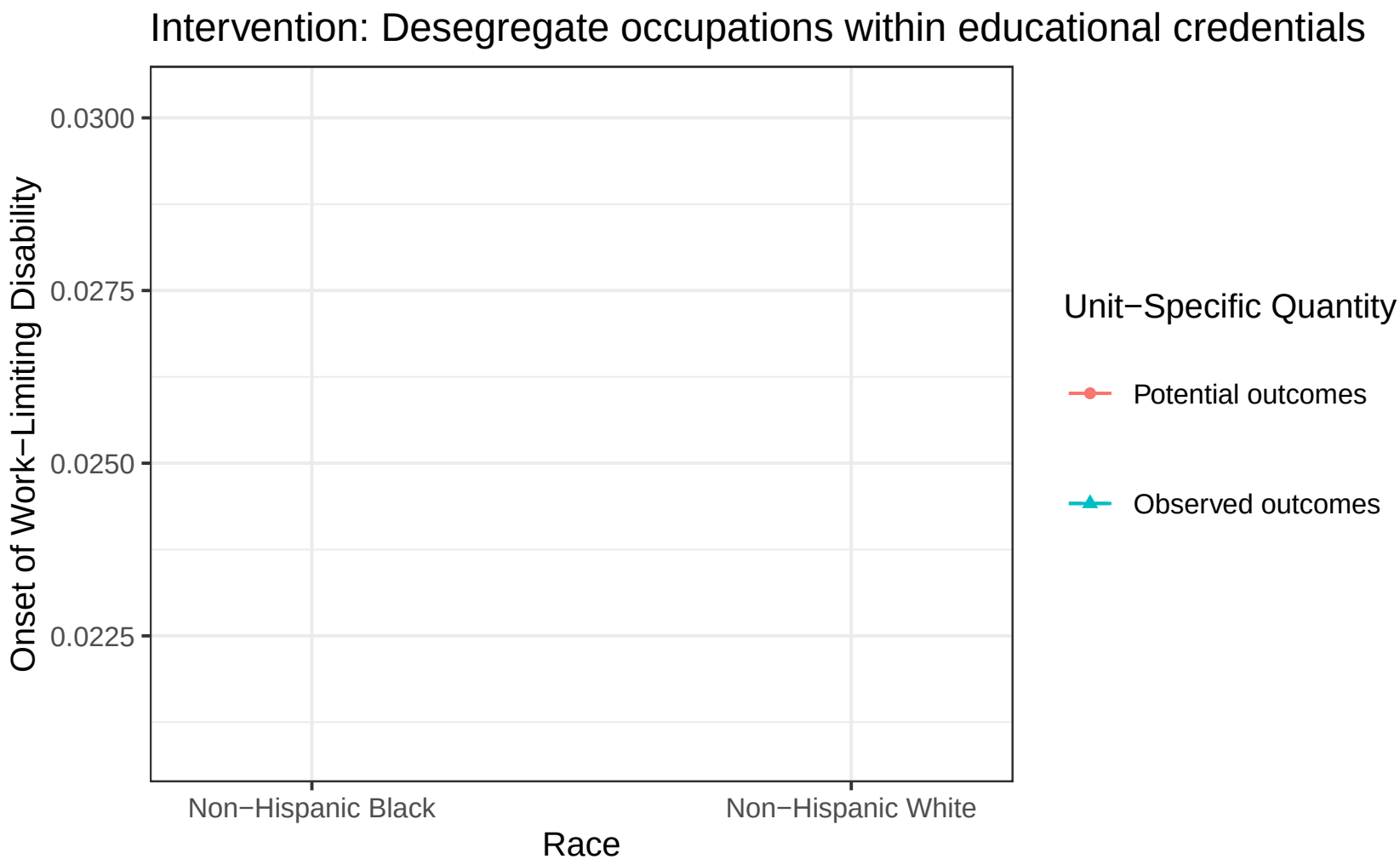
Sex,
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Foreign born,
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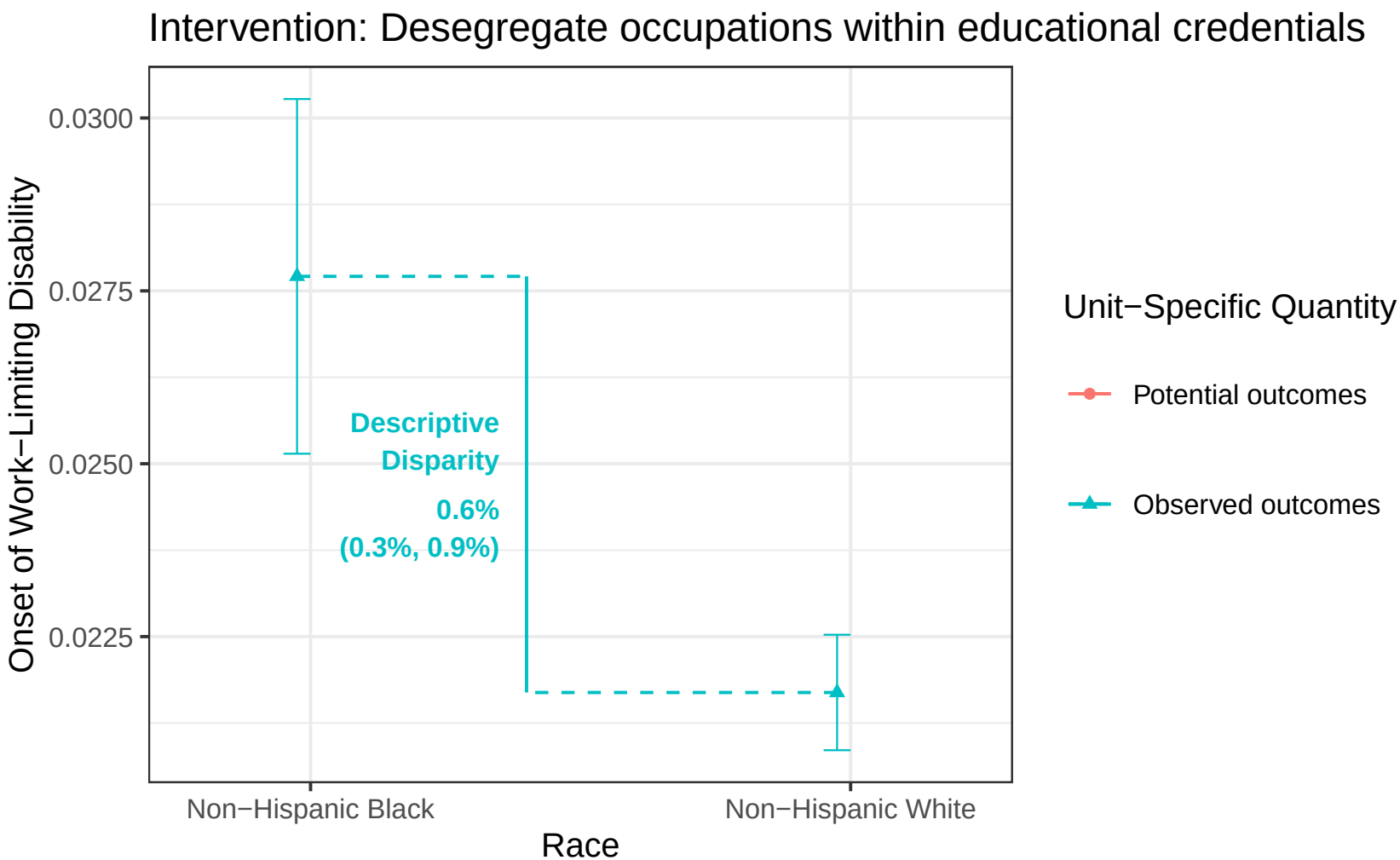


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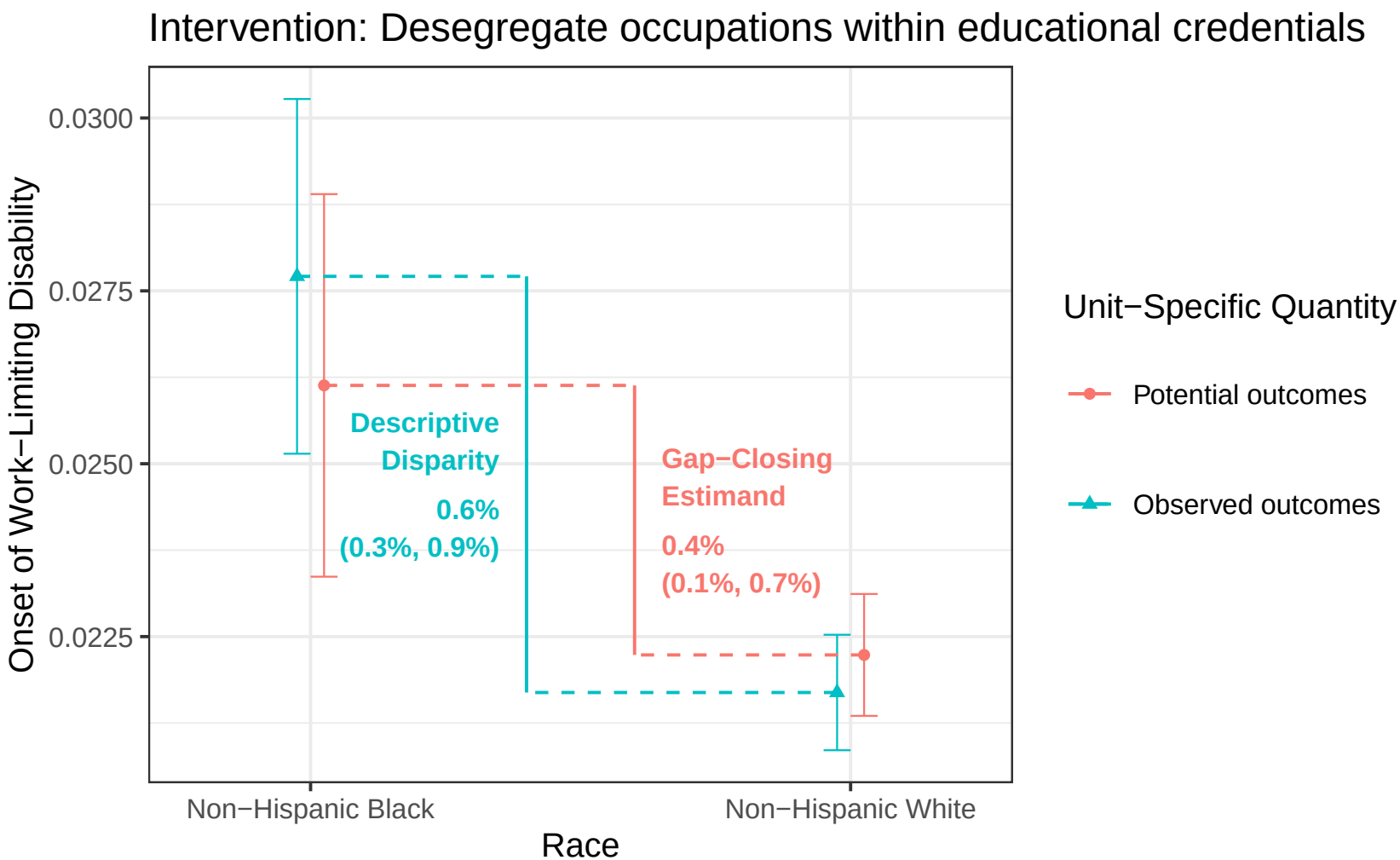
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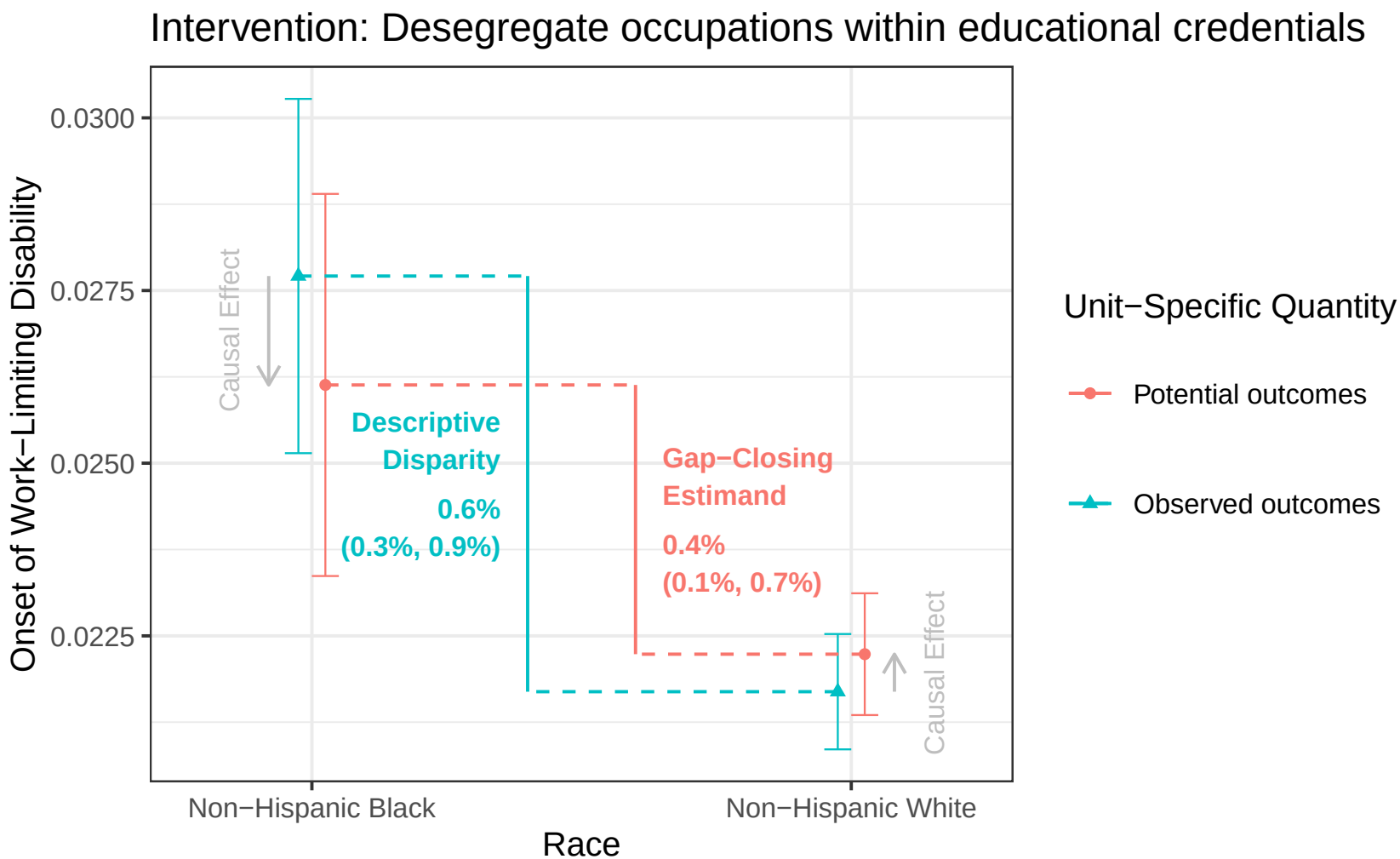
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The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

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- Produce software
- **Impact: Others using these ideas**

Impact: We can now study gaps in a **new way**

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The Black-white gap would close by more than one-fourth

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SAGE

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The gap would close by more than half for men, but not for women

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CHILD DEVELOPMENT

EMPIRICAL ARTICLE

Environmental inequality and disparities in school readiness: The role of neurotoxic lead

Jared N. Schachner[1] | Geoffrey T. Wodtke[2]

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$$\mathrm{E}(Y^1 - Y^0)$$

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Effect on median	$\mathrm{Median}(Y^1) - \mathrm{Median}(Y^0)$
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Gap-closing for variance	$\mathrm{Var}(Y^1 \mid X = 1) - \mathrm{Var}(Y^1 \mid X = 0)$

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Effect on variance	$\mathrm{Var}(Y^1) - \mathrm{Var}(Y^0)$
Gap-closing for variance	$\mathrm{Var}(Y^1 \mid X = 1) - \mathrm{Var}(Y^1 \mid X = 0)$
Counterfactual slopes	$\frac{\mathrm{Cov}(Y^1, X)}{\mathrm{Var}(X)}$ for continuous X (Yu & Zhao 2024)

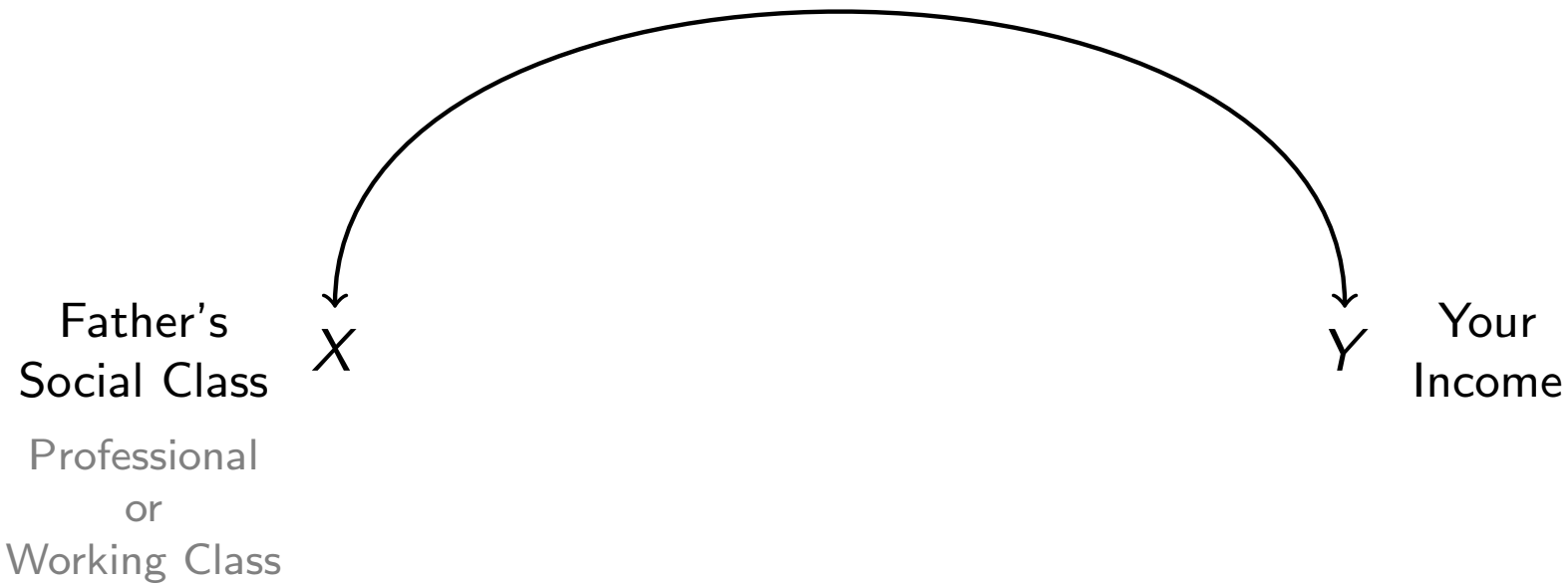
Thanks!

Ian Lundberg
ianlundberg.org
ianlundberg@ucla.edu

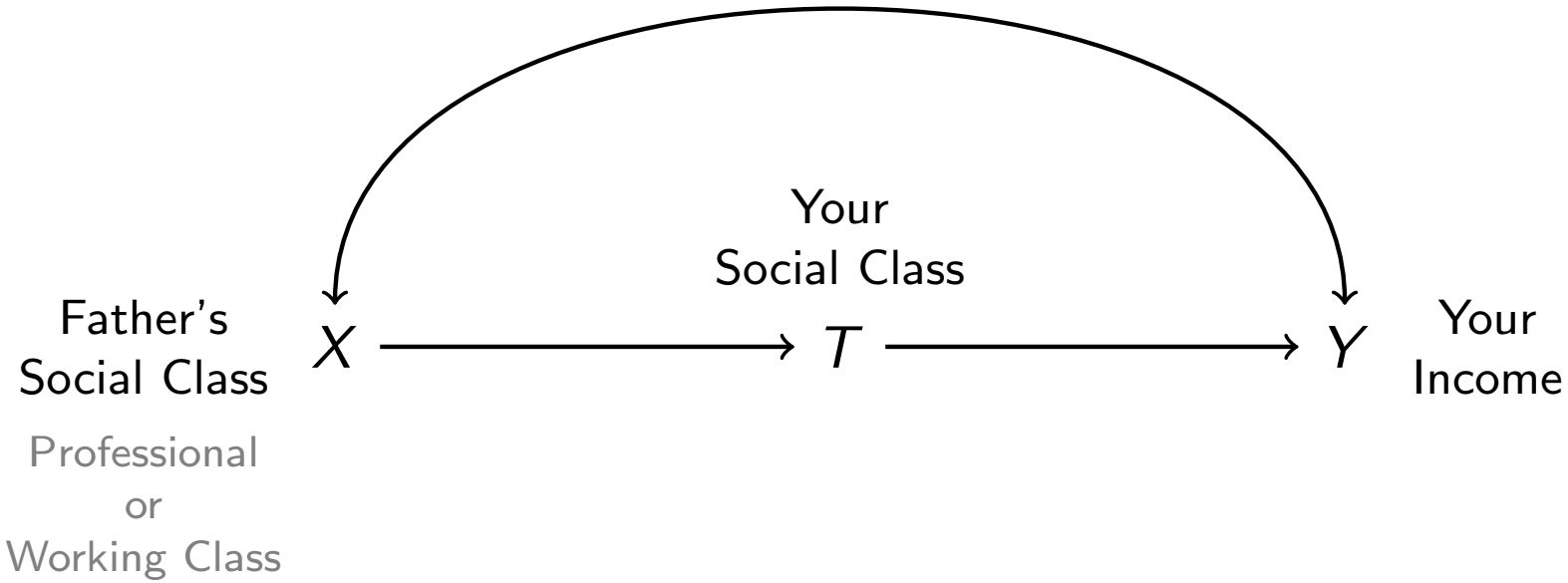
More about this project with a code tutorial:
ilundberg.github.io/gapclosing

Appendix

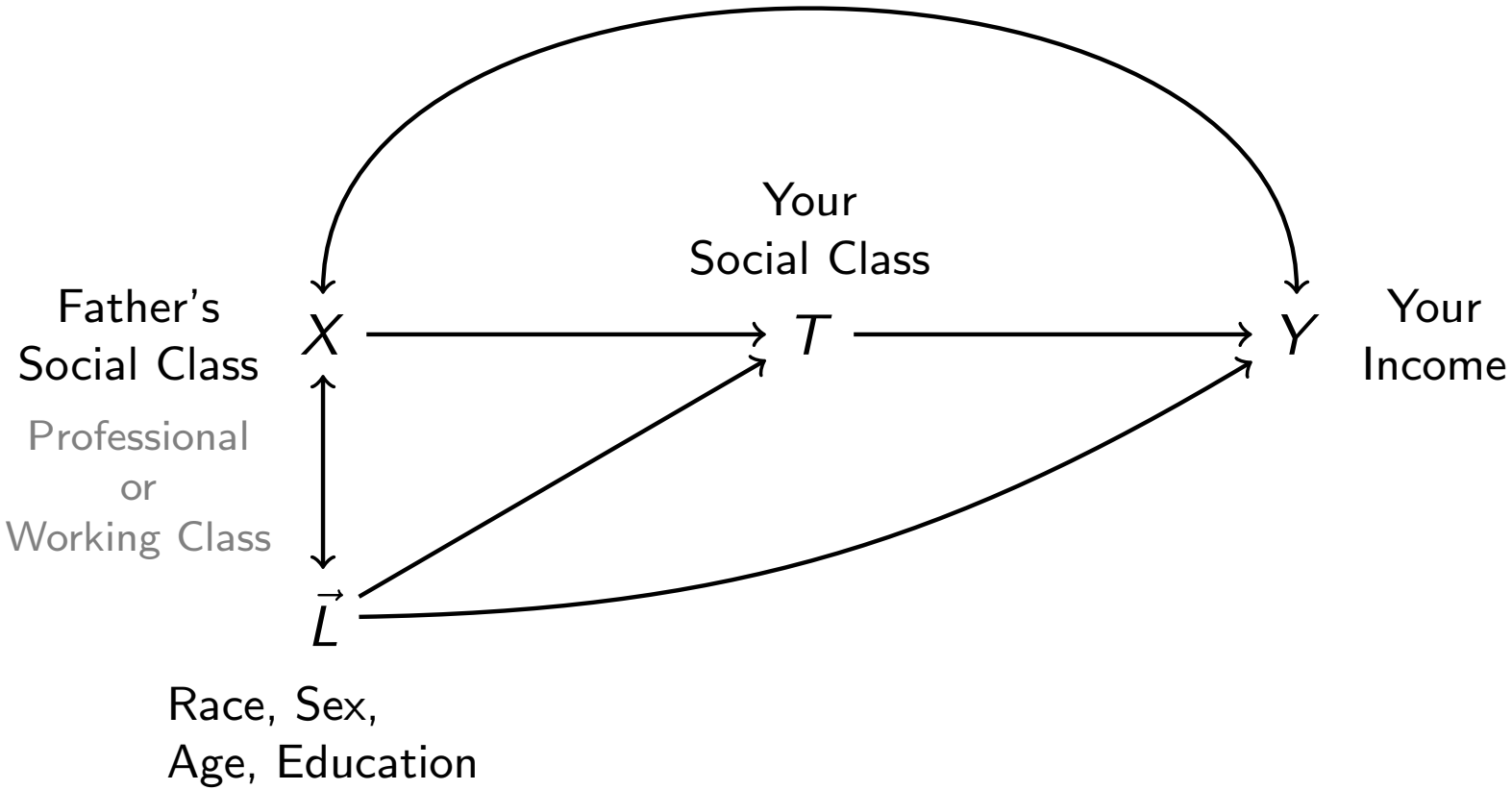
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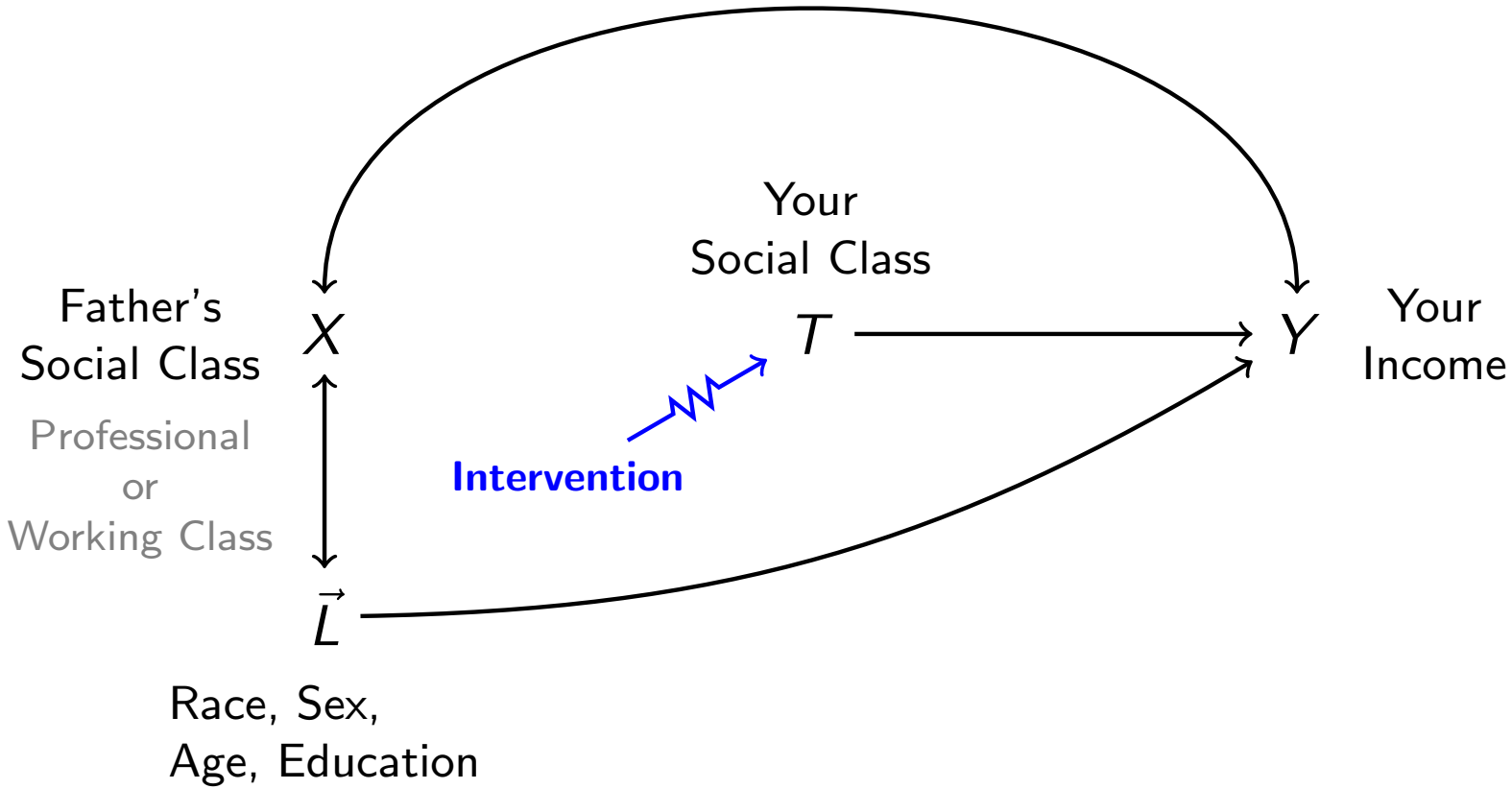
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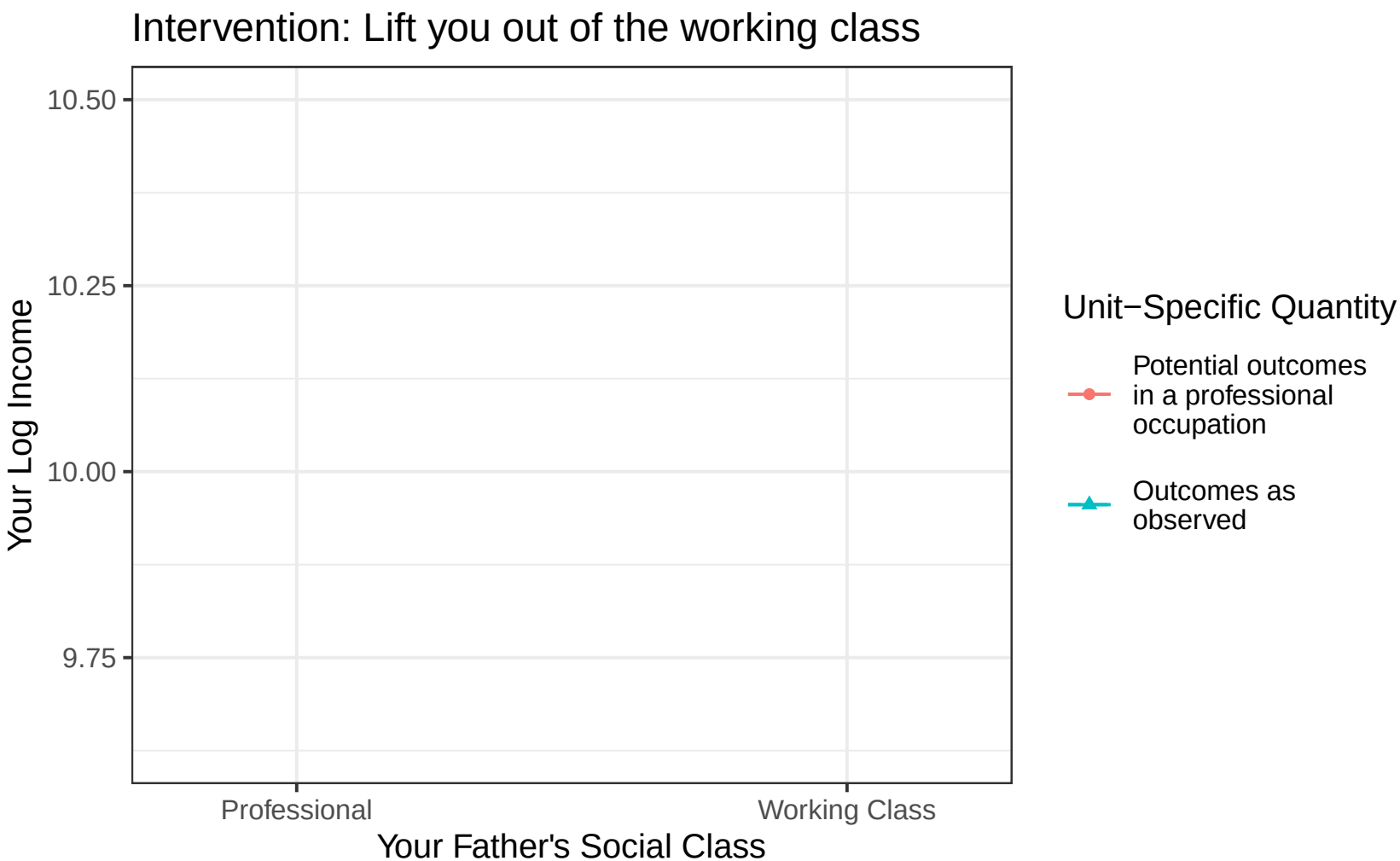
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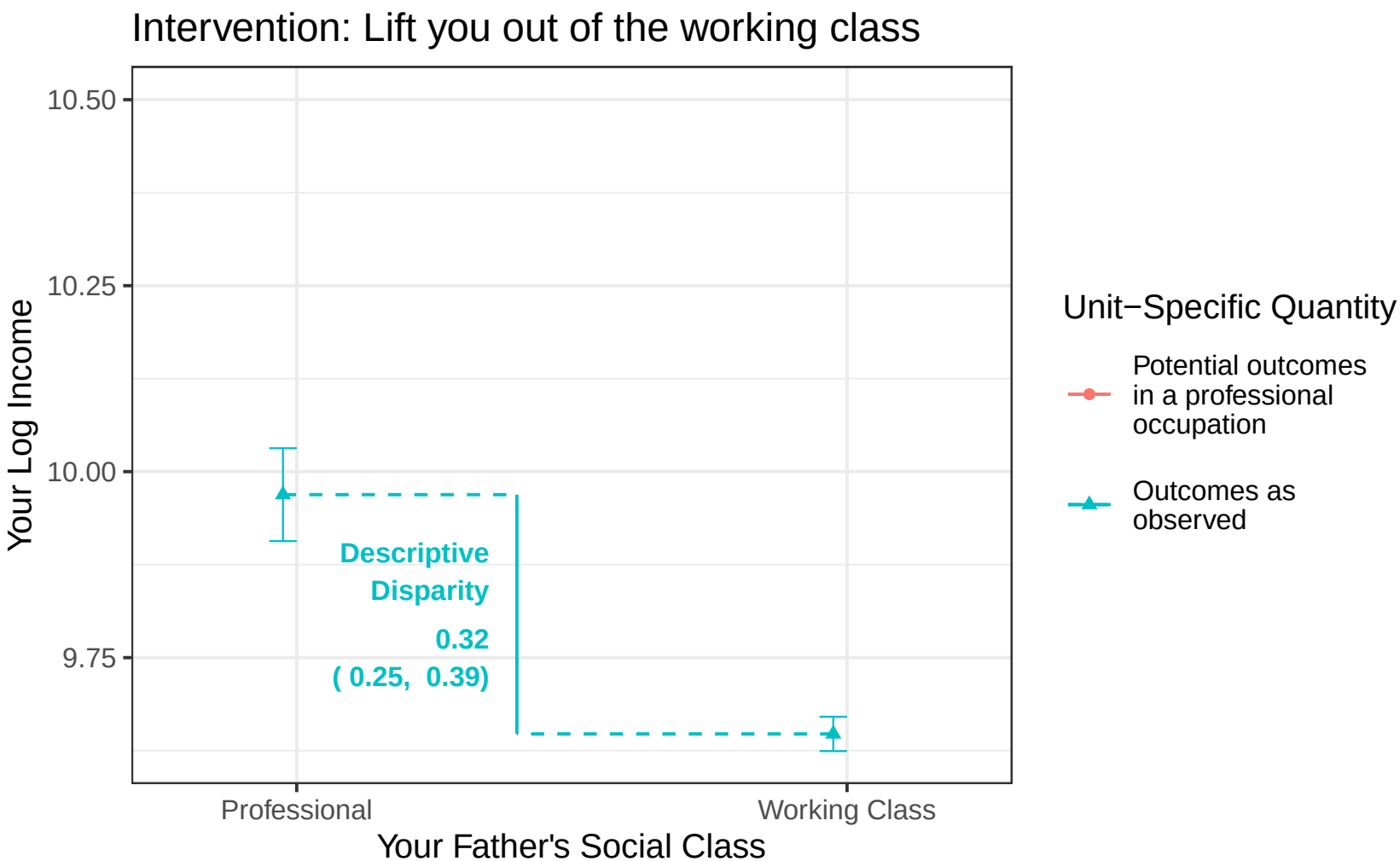
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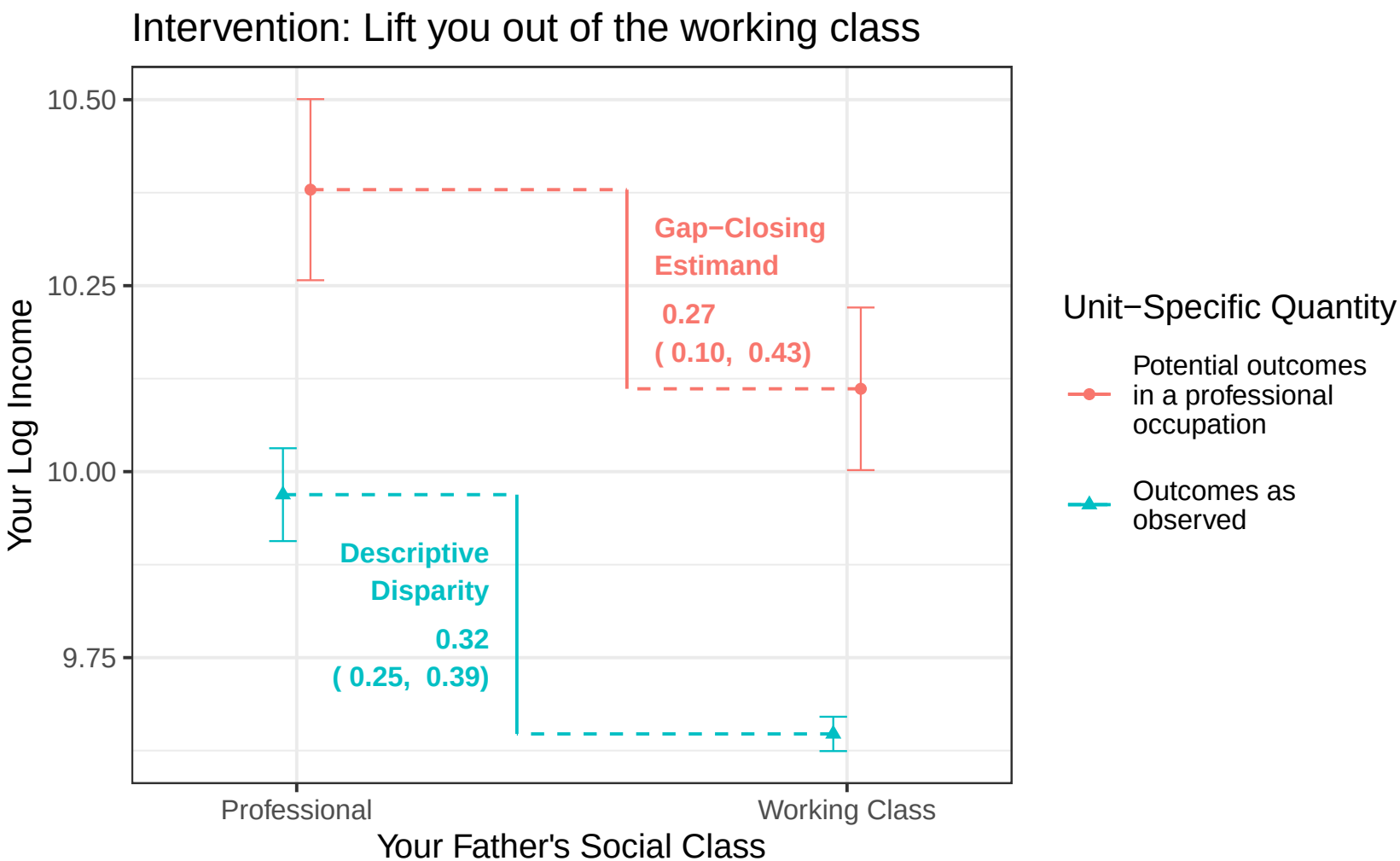
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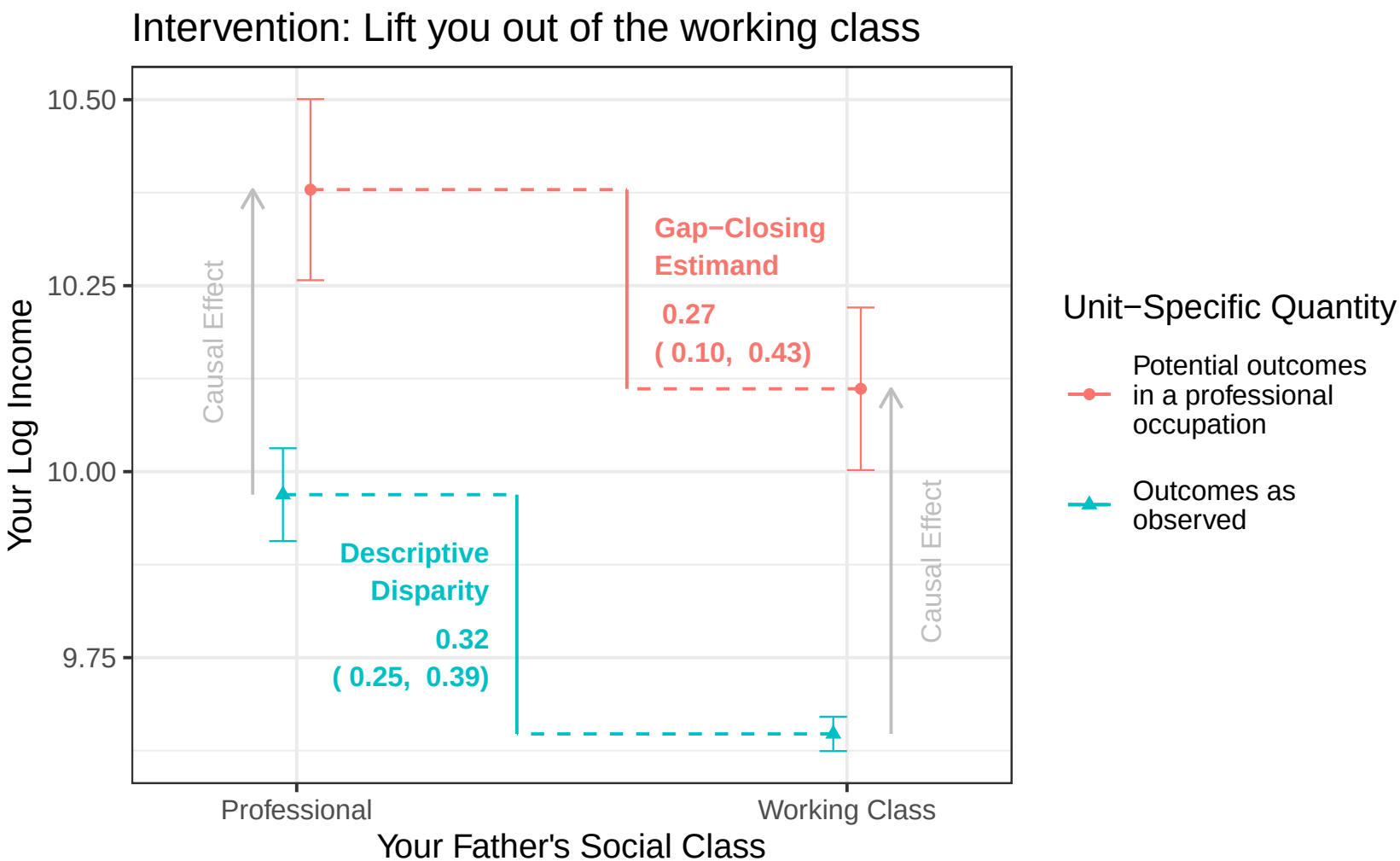
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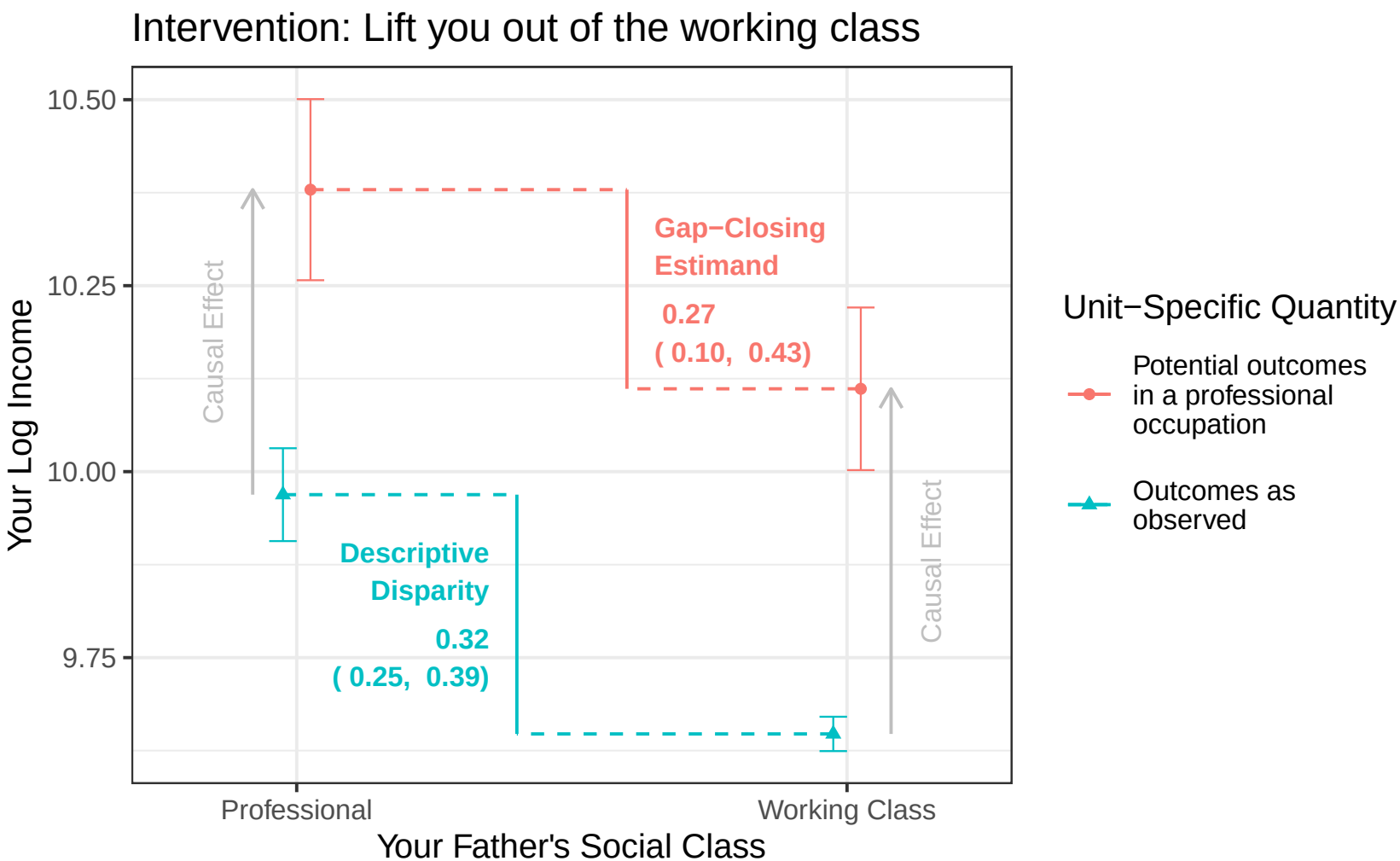


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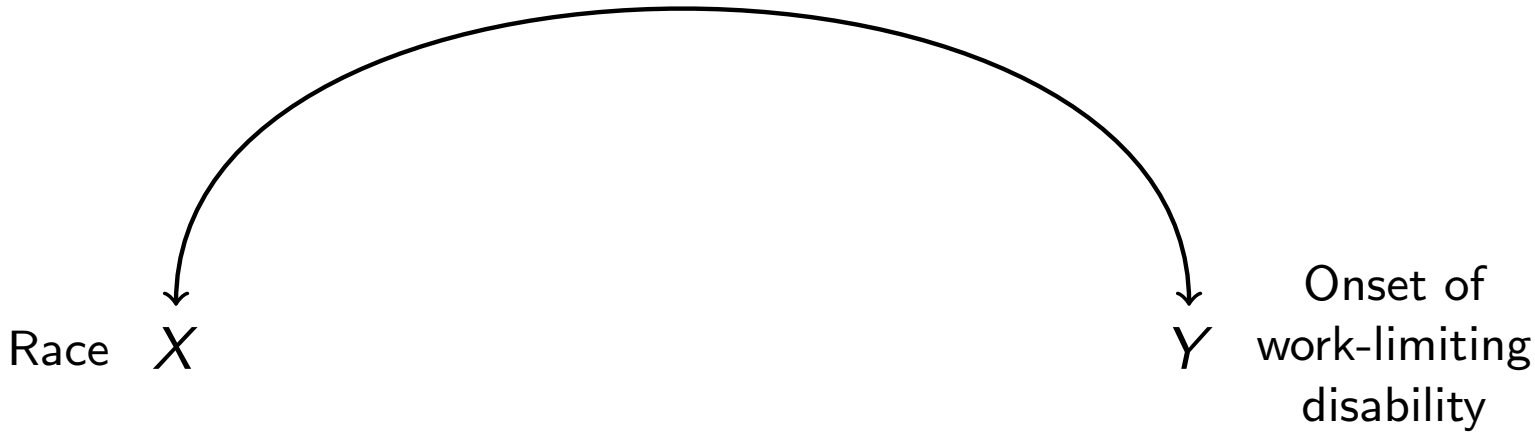


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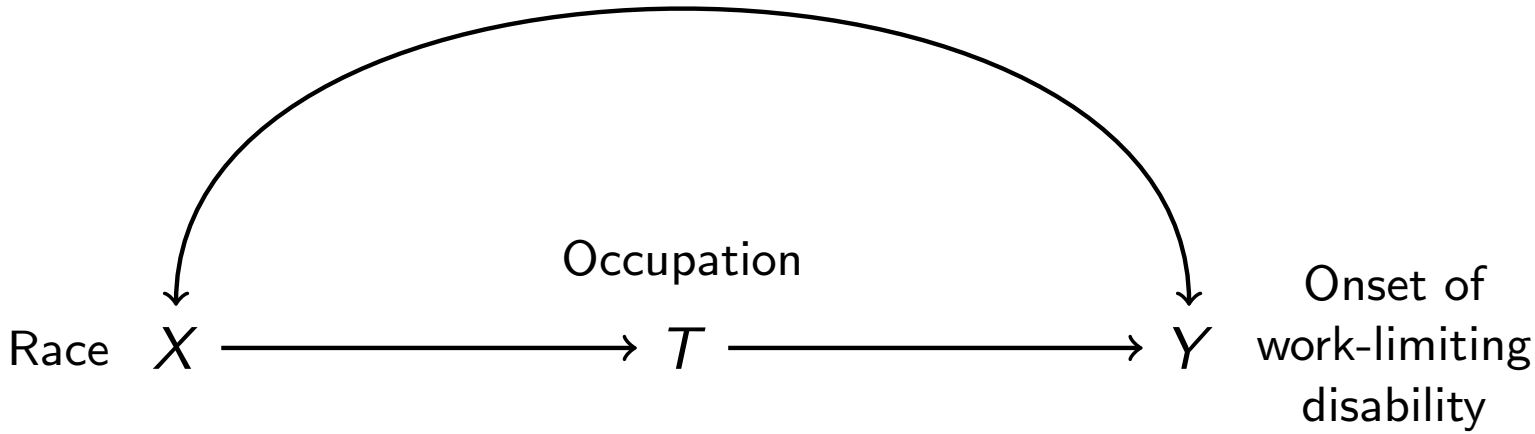
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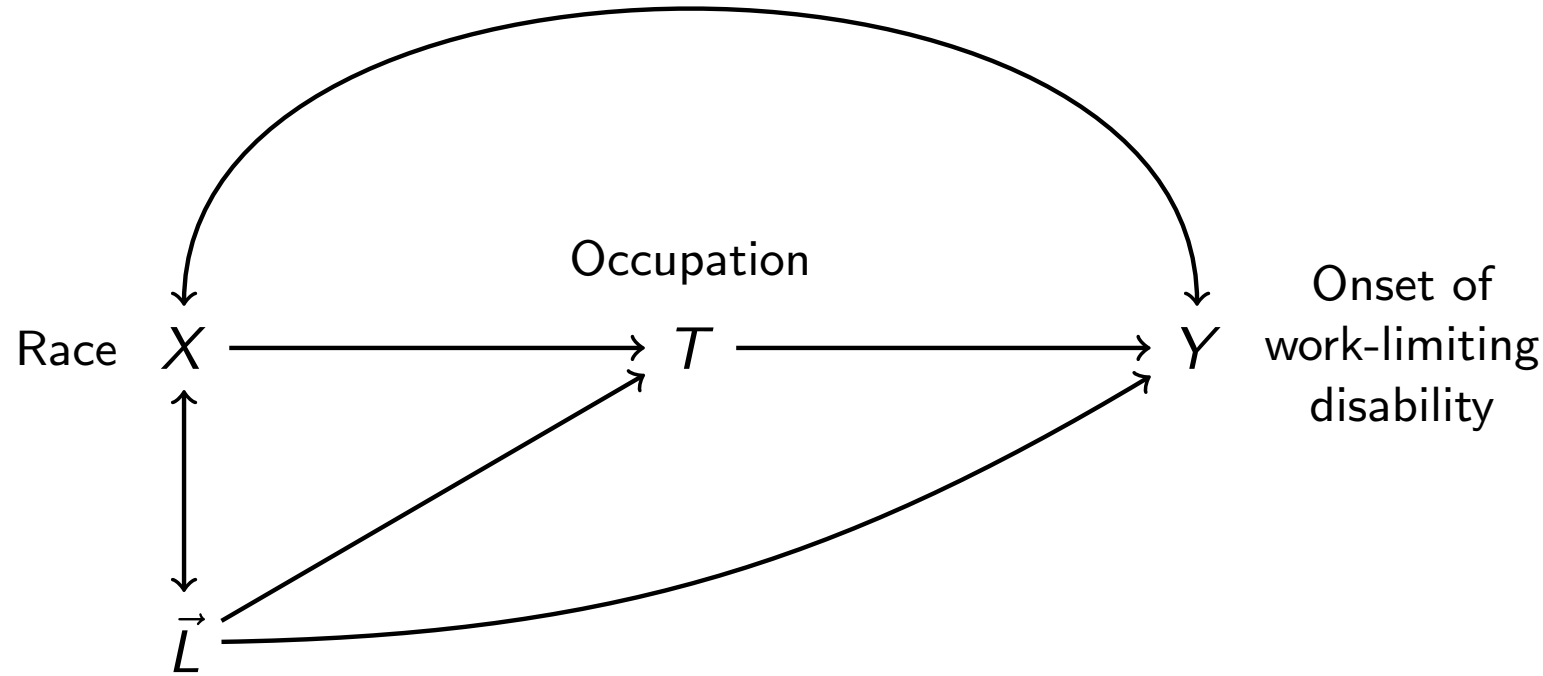
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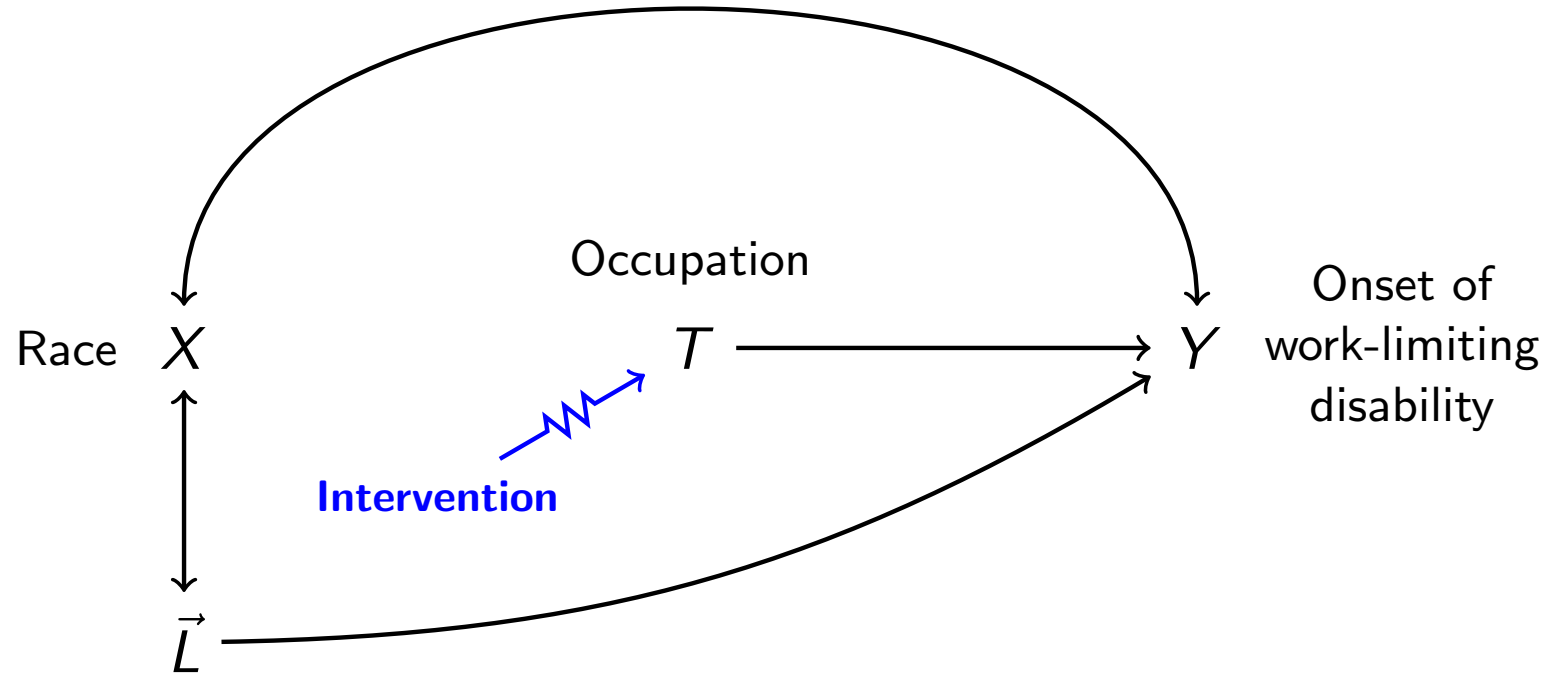


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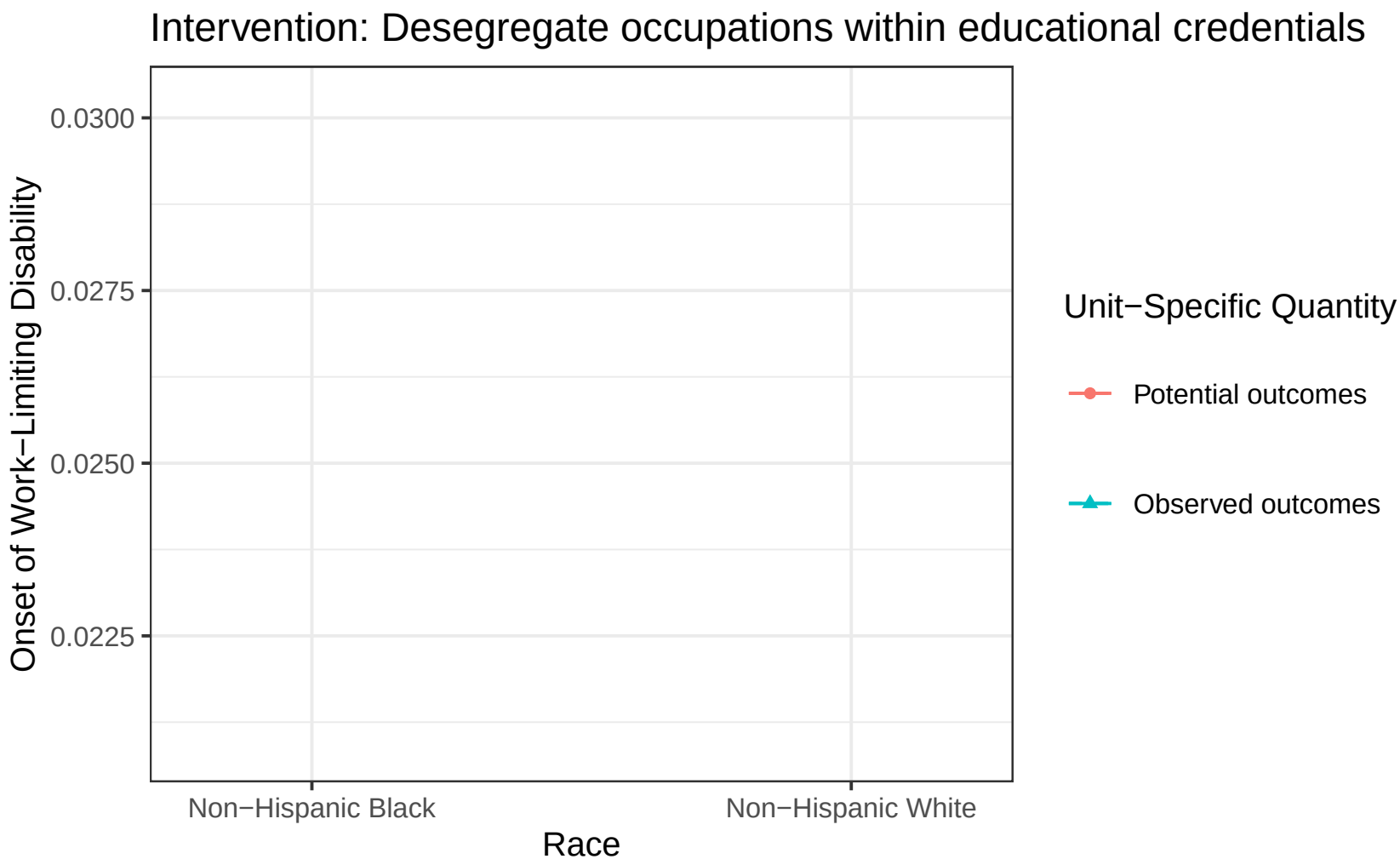
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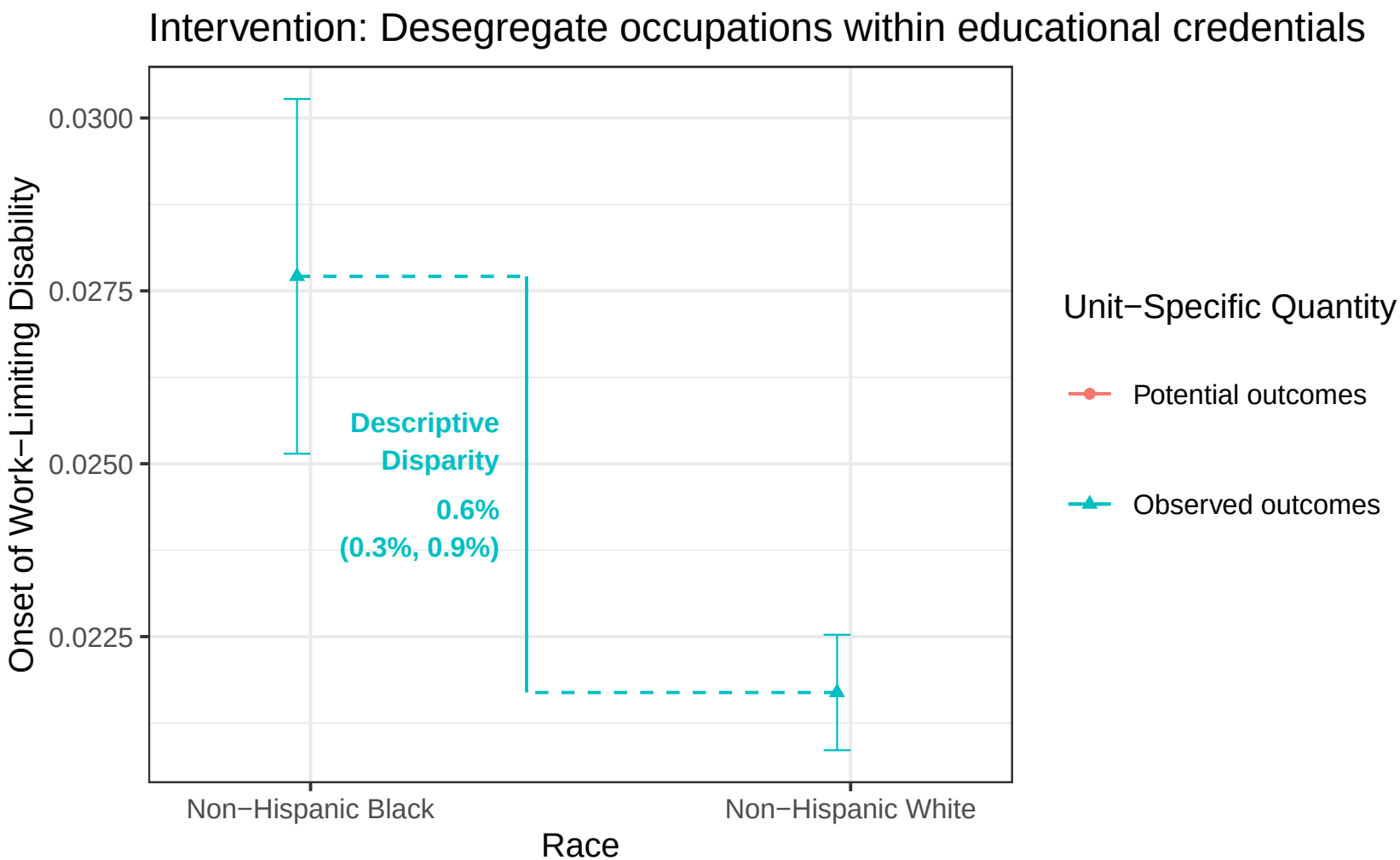


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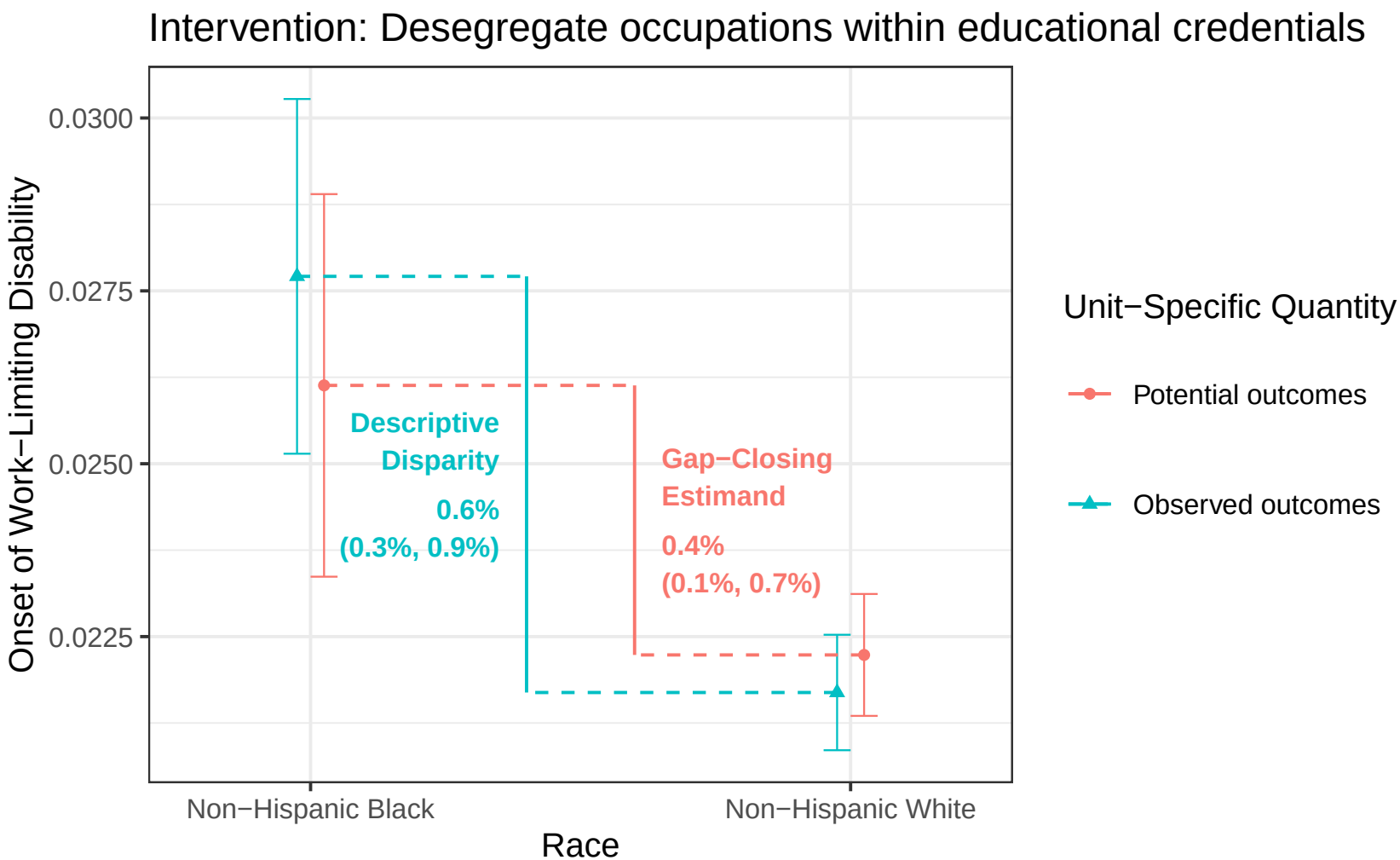
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